Spending Smarter under Demographic Pressure:

Pension Reform, Health, and Household Welfare in Mexico*

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Abstract

This paper examines how demographic aging affects pension financing, macroeconomic outcomes, and household welfare in Mexico. We use a stochastic overlapping-generations general-equilibrium model with endogenous health investment and heterogeneity by sex and education, calibrated and estimated in two stages on Mexican data. We compare stationary equilibria calibrated to Mexico's 2020 old-age dependency ratio and to its projected 2050 value. If the replacement rate and the retirement age remain unchanged, the pay-as-you-go contribution rate rises from 5.66% to 12.26%. Capital deepening raises wages and lowers the interest rate, leaving aggregate newborn welfare nearly unchanged relative to the 2020 baseline, while welfare comparisons become increasingly negative for older residents. Relative to this contribution-rate adjustment, reducing the

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replacement rate from 40% to about 18.5% raises newborn welfare by 8.29% of lifetime consumption, and raising the retirement age from 65 to 70 raises it by 3.11%. The reforms differ in their effects across ages, sex, and education, particularly for households that rely more on pension income. The model's health block reproduces only a small part of the observed education gap in survival, a result we relate to the predominance of curative spending in Mexican out-of-pocket health expenditure. All results compare stationary economies and do not include the transition costs of implementing the reforms.

Keywords: demographic transition; fiscal sustainability; social security; pension reform; health capital; overlapping generations; Latin America
JEL codes: E62, H55, I13, J11, J26

1 Introduction

Mexico's old-age dependency ratio is projected to more than double between 2020 and 2050, from 0.141 to 0.307 (CONAPO, 2023). In a balanced pay-as-you-go pension system, an increase in the ratio of retirees to working-age individuals requires higher payroll contributions if the replacement rate and retirement age remain unchanged. Alternatively, the adjustment can be accommodated through lower pension benefits or later retirement. These alternatives have different implications for household saving, labor supply, and the distribution of resources across generations, while also affecting equilibrium wages and interest rates. In this paper, we quantify the macroeconomic and welfare consequences of demographic aging under these alternative pension adjustments in Mexico.

To examine these interactions, we develop a quantitative stochastic overlapping-generations general-equilibrium model calibrated to the Mexican economy. Households differ by sex, education, and idiosyncratic productivity, and make life-cycle decisions about consumption, saving, labor supply, and medical spending. Medical spending affects health, which in turn influences survival probabilities, labor productivity, and utility. Household saving and labor supply decisions determine aggregate capital and effective labor, and hence equilibrium wages and interest rates. The model incorporates a pay-as-you-go pension system, allowing us to examine how alternative pension adjustments affect household decisions and welfare through changes in pension benefits, payroll contributions, and equilibrium factor prices. The pension block is a stylized pay-as-you-go benchmark rather than a representation of the full Mexican system, which also includes individual accounts (Section 3.5).

We discipline the model using Mexican data on life-cycle earnings, medical spending, and survival. The National Survey of Household Income and Expenditure (ENIGH) provides earnings profiles by age, sex, and education, as well as information on how medical spending varies with age and income. The Mexican Health and Aging

Study (ENASEM) provides evidence on health deterioration and mortality differences across demographic groups. We combine these data with demographic, production, and fiscal parameters obtained from official statistics and existing research.

Using the calibrated model, we first examine the effects of demographic aging when the pension system adjusts through higher payroll contributions, holding the replacement rate and retirement age unchanged. The contribution rate increases from 5.66% to 12.26%. At the same time, demographic aging leads to capital deepening, raising wages by 12.4% and reducing the annual interest rate from 8.58% to 6.82%. These changes have offsetting effects on household welfare, leaving aggregate newborn welfare nearly unchanged, with a consumption-equivalent gain of 0.14% relative to the 2020 baseline. However, the age-specific welfare comparisons reveal substantial differences. Relative to the 2020 equilibrium, residents of the aged stationary economy experience increasingly large welfare losses at older ages, reaching 12–21% of remaining lifetime consumption at ages 65–69.

We next examine two alternative pension reforms under the same demographic conditions. Reducing the pension replacement rate from 0.40 to approximately 0.185 restores the contribution rate to its 2020 level and increases aggregate newborn welfare by 8.29% relative to contribution-rate adjustment. Raising the retirement age from 65 to 70 reduces the contribution rate to 8.78% and generates a newborn welfare gain of 3.11% relative to contribution-rate adjustment. The benefit reduction encourages private saving and generates additional capital deepening, raising wages and lowering interest rates. In contrast, raising the retirement age increases capital and effective labor in similar proportions, leaving factor prices nearly unchanged.

The welfare consequences of these adjustments also differ substantially by sex and education, particularly at older ages. Under contribution-rate adjustment, welfare losses among older residents are greatest for high-education households, which hold more assets and are more exposed to the decline in interest rates. In contrast, reducing pension benefits generates larger losses among households that rely more heavily on pension income. For example, at ages 65–69, low-education women experience welfare losses equivalent to approximately 21% of remaining lifetime consumption, compared with 3.7% for high-education men. Raising the retirement age also disproportionately affects low-education women, whose welfare at ages 65–69 declines by 9.5%, while high-education individuals experience welfare gains across all ages. These findings highlight the importance of differences in life-cycle earnings, asset accumulation, and survival across sex and education groups for evaluating the distributional consequences of pension reform.

Our paper contributes to the quantitative literature on demographic aging and pension reform by examining their welfare consequences in Mexico, with particular attention to differences across age, sex, and education. Previous studies have analyzed the general-equilibrium effects of demographic aging and alternative pension policies (Krueger and Ludwig, 2007; Kitao, 2014). We extend this analysis to a model

calibrated to the Mexican economy, in which the pension contribution rate adjusts endogenously to demographic change and households differ in their life-cycle earnings and survival probabilities. These differences allow us to examine how changes in pension benefits, contributions, and returns on savings affect the welfare of different demographic groups. Our framework also incorporates endogenous medical spending, through which health influences labor productivity and longevity. Related overlapping-generations models have studied endogenous health in general equilibrium (Jung et al., 2017), while Cortés et al. (2024) examine health heterogeneity in Mexico in partial equilibrium. We bring these elements together to evaluate the aggregate and distributional consequences of alternative pension adjustments in a common quantitative framework. The estimated health block reproduces only a small part of the observed education gap in survival, generating a gap of 0.30 percentage points in five-year survival at ages 50–64 against 2.09 in the data. Section 4.8 shows that health remains close to its upper bound in the calibrated equilibrium, and Appendix D documents that most out-of-pocket medical spending in Mexico is curative, whereas the model treats medical spending as chosen before the health shock is realized.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model, and Section 4 describes its calibration and evaluates its quantitative performance. Section 5 presents the counterfactual experiments, and Section 6 concludes.

2 Literature Review

This paper draws on four literatures: quantitative overlapping-generations models of demographic aging and pension reform; life-cycle models with endogenous health capital; studies of health financing and social insurance in Latin America; and work on heterogeneity by sex and education in life-cycle fiscal models. The review concentrates on the first, second, and fourth.

2.1 Demographic Aging and Pension Reform

Since Auerbach and Kotlikoff (1987), the multi-period overlapping-generations model with a government sector has been the standard tool for studying the fiscal consequences of demographic change. Krueger and Ludwig (2007) show that population aging raises the capital–labor ratio, so wages rise and the return on capital falls. The welfare consequences then depend on the composition of a household's resources. Households whose lifetime resources consist mainly of labor income gain from the higher wage, while households that already hold assets lose from the lower return. Krueger and Ludwig (2007) also show that the adjustment of pension contributions can offset part of the direct effect of aging. Carvalho et al. (2016) separate the longevity channel from the dependency-ratio channel in the determination of the equilibrium

return; the experiments in this paper change population growth only and hold survival schedules fixed. Auclert et al. (2021) quantify these forces across countries and find that the sign and size of the aging effect depend on country characteristics. This paper studies the same channels in a Mexican calibration in which the pay-as-you-go contribution rate adjusts endogenously to balance the pension budget.

Kitao (2014) compares four adjustments that restore the balance of a pay-as-you-go system under aging: a higher payroll tax, a lower replacement rate, a higher retirement age, and means-tested benefits. This paper considers the first three. The flat benefit of Section 3 cannot represent means-testing, so that option is omitted because of the benefit rule rather than a judgment about the instrument. Huggett and Ventura (1999) compare pension arrangements across stationary economies and show that the choice among them is largely distributional; their exercise is the precedent for the comparison of steady states in this paper. Conesa and Krueger (1999) and Nishiyama and Smetters (2007) show that an unfunded pillar insures households against uninsurable idiosyncratic wage risk, and Nishiyama and Smetters (2007) find that the efficiency gains from reducing it shrink or reverse once transitional generations are compensated. Newborn welfare gains measured across steady states therefore provide an upper bound on the value of a reform once the transition is included.

Differences in mortality by socioeconomic status also affect the incidence of pension systems. Sánchez-Romero et al. (2020) show that longevity gaps by socioeconomic status transfer resources toward the long-lived and reduce the actuarial fairness of pension systems. Bagchi (2019) finds that optimal pension progressivity is insensitive to wealth-dependent mortality. This paper reports the incidence of each adjustment by education group rather than an optimal policy.

The welfare comparisons rest on the condition stated by Aaron (1966). Where the market return exceeds the growth rate of the contribution base, mandatory participation in an unfunded pillar transfers resources away from participants. Abel et al. (1989) show that this comparison is more delicate under uncertainty. In the deterministic aggregate environment of this model, the annual return ranges from 5.89 to 8.58 percent against base growth of at most 3.51 percent, so the condition holds in every economy considered. The factor-price channel is a closed-economy object. Attanasio et al. (2007) show that capital mobility between regions aging at different rates mitigates the factor-price effects of aging, so the closed-economy assumption matters for the factor-price results reported below. Pessino and Ter-Minassian (2021) project the pension and health spending pressure that aging implies for Latin America over the coming decades.

2.2 Health Capital and the Life Cycle

The health block follows Grossman (1972). Health is a depreciable capital stock, medical spending is investment in that stock, and the stock affects survival and, in this paper,

labor productivity. Several models build on this structure. Dalgaard and Strulik (2014) derive aging and death from the interaction of health investment with biological depreciation. Kuhn and Feichtinger (2015) analyze health and retirement decisions jointly when health capital depreciates with age and can be partially restored. Fioroni (2010) shows that the interaction of precautionary saving with health investment produces consumption and saving profiles that differ from those of models with exogenous health. Hall and Jones (2007) show that when the marginal value of additional life rises with consumption, the income elasticity of health spending exceeds one and the expenditure share of health grows with development, and Murphy and Topel (2006) develop the same valuation structure and its empirical magnitudes. The multiplicative health amenity in this paper, and the income gradient in the willingness to pay for health that it generates, follow these two papers.

Jung and Tran (2016) and Jung et al. (2017) build stochastic general-equilibrium overlapping-generations economies with endogenous health capital and health insurance, and Jung et al. (2017) use that model for aging experiments for the United States. Their model is the closest to the one in this paper. Zhao (2014) shows that social security itself alters health investment and longevity. Franković and Wrzaczek (2020) trace medical progress through capital accumulation and factor prices, Yew and Zhang (2018) introduce longevity externalities, Lim (2020) shows that public health insurance changes aggregate saving in Korea, and Carrasco (2026) models health investment as a channel through which longer lives raise life-cycle saving and move the equilibrium interest rate. A separate strand treats health as a source of expenditure risk. De Nardi et al. (2010) show that uncertain medical expenses are a primary driver of late-life saving among older American households. Hosseini et al. (2022) estimate a health-deficit process on United States panel data in which health is observed from early adulthood. Gertler and Gruber (2002) show for a developing economy that major illness is a large and poorly insured shock to household consumption. In this model, the shocks hit the health stock rather than the household budget.

These models are calibrated or estimated mostly on high-income data, in economies where coverage is broad and out-of-pocket payments are a small share of health spending. Such data do not separate spending that builds the health stock from spending that restores it after a shock. Ozkan (2025) separates preventive from curative spending in a life-cycle health-capital model on United States data and documents how their income gradients differ over the life cycle. Galama and van Kippersluis (2013) discuss reformulations of the Grossman model that address the standard criticisms of the framework. For Mexico, Cortés et al. (2024) develop a partial-equilibrium overlapping-generations model in which agents differ by education and face stochastic health transitions, health-dependent productivity, and age-varying out-of-pocket expenditure. The present paper combines endogenous health capital, heterogeneity by sex and education, and an endogenously balanced pay-as-you-go system in a general-equilibrium model calibrated to Mexico and disciplined against its own ENIGH and ENASEM mo-

ments. Medical spending in this model is chosen before the health shock is realized, so it is preventive in timing, while most out-of-pocket medical spending in Mexico is curative. Section 4.8 and Appendix D examine the consequences of this difference.

2.3 Health Financing and Social Insurance in Latin America

Health shocks are large and poorly insurable through private markets, so pooling arrangements can reduce the associated welfare losses (Arrow, 1963). Most Latin American countries operate hybrids of Bismarckian systems, which link coverage to formal employment, and Beveridgean systems, which link it to residency, often with fragmentation and unequal benefit packages (Levy, 2008). Summers (1989) shows that mandated contributions act as a tax on formal labor when workers do not value the benefits they finance. Fietz et al. (2025) measure this logic as a formalization tax rate, the net fiscal cost of formal employment inclusive of forgone non-contributory benefits, and find that it is highest for low-income workers. Baeza and Packard (2006) document that workers value health coverage, which protects against immediate expenditure risk, more than pension benefits that arrive far in the future. In this model, pension benefits are a flat transfer indexed to average earnings and financed by payroll contributions, with no link between individual contributions and benefits and no formal/informal margin. The results therefore concern pension financing under a stylized pay-as-you-go system rather than formalization incentives, although the contribution rate here is determined by the system's budget rather than imposed.

2.4 Heterogeneity by Sex and Education

Estimated life-cycle models increasingly study how social security affects men and women differently. Borella et al. (2023) estimate a model of couples and singles with survival risk, medical expenses, and the spousal provisions of United States Social Security, and show that those provisions have large and unequal effects on women's labor supply and welfare. Fehr and Kindermann (2018) model a two-person household in which female labor supply responds to childcare, parental leave, and joint versus individual taxation, in an environment without health or endogenous survival. Combining such a household with continuous health capital, health-dependent survival, and endogenous factor prices would raise the computational burden considerably. This paper instead uses four types, two education groups for each sex, each with its own measured age-efficiency profile and baseline survival schedule. These schedules capture in reduced form the differences in labor-market histories and mortality that a structural household model would generate endogenously. Welfare results are reported for all four groups in Section 5. A structural household model remains a natural extension.

Section 3 presents the model, which extends Cortés et al. (2024) to general-

equilibrium factor prices, a continuous health stock with age-dependent depreciation, and an endogenously balanced pay-as-you-go system. Section 4 describes the calibration and Section 5 the experiments.

3 Model

3.1 Environment and Demographics

The economy consists of J overlapping generations, indexed by $j = 1, \ldots, J$, with each period representing five years. Households enter the economy at age 20 ($j = 1$), retire at the exogenously specified retirement age j_R, and may live until age 100 ($j = J$). The baseline is calibrated to the formal-sector population, and the dependency ratio that closes the pension system is demographic rather than contributor-based.

Agent types. Individuals differ by sex $g \in \{m, f\}$ (male, female) and permanent skill type $\theta \in \{\theta_L, \theta_H\}$ (low-skilled, high-skilled). The two skill types correspond to education groups, and we use the terms interchangeably. Both characteristics are fixed at birth. Labor productivity depends on the skill-specific log-productivity intercept θ and a sex- and skill-specific age-efficiency profile $e_j^{g,\theta}$, which allows earnings to evolve differently over the life cycle across groups (equation (12)). Health also affects productivity, with a common coefficient ϱ across groups. Survival probabilities depend on age, sex, education, and health. The baseline survival probabilities $\psi_j^{\circ\, g,\theta}$ capture differences in mortality across groups that are not generated by endogenous health investment (equation (11)). Let $\pi_1^{g,\theta}$ denote the share of individuals entering the economy at $j = 1$ who belong to type (g, θ), with $\sum_{g,\theta} \pi_1^{g,\theta} = 1$. Population grows at gross rate $1 + n_p$ per model period.

Idiosyncratic risk. Households face three sources of uncertainty: mortality risk, idiosyncratic labor-productivity risk, and health shocks. Survival probabilities depend on health, while the idiosyncratic component of labor productivity, η, follows an AR(1) process. Health shocks occur with age-dependent probability and reduce health by a fixed amount, subject to a lower bound (Section 3.4.3). Households influence their health through medical spending, but health remains subject to stochastic shocks. Productivity and health shocks are independent, and there is no aggregate uncertainty.

Individual state. At the beginning of each period, an individual's state is

$$s = (a,\ h,\ \eta), \tag{1}$$

where $a \in \mathcal{A} = [0, \bar{a}]$ denotes beginning-of-period assets, $h \in \mathcal{H}_h = (0, \bar{h}]$ denotes health capital, and $\eta \in \mathcal{H}_\eta = \{\eta_1, \ldots, \eta_{N_\eta}\}$ denotes the idiosyncratic component of

labor productivity. The value and policy functions are also indexed by age j, sex g, and skill type θ.

Cohort weights. Let $\phi_j^g(\theta)$ denote the share of the stationary population of age j, sex g, and skill θ. The shares of individuals entering the economy at age $j = 1$ satisfy:

$$\phi_1^g(\theta) \propto \pi_1^{g,\theta},$$

where $\pi_1^{g,\theta}$ is the share of type (g, θ) among entrants. The stationary population shares evolve according to

$$\phi_{j+1}^g(\theta) \;=\; \frac{\bar{\psi}_{j+1}^g(\theta)}{1+n_p}\,\phi_j^g(\theta), \qquad j = 1, \ldots, J-1, \tag{2}$$

where $\bar{\psi}_{j+1}^g(\theta)$ is the average probability of surviving from age j to age $j+1$ among individuals of sex g and skill θ. This probability is given by

$$\bar{\psi}_{j+1}^g(\theta) \;=\; \frac{1}{\phi_j^g(\theta)} \int \sum_{k\in\{0,1\}} \Pr(k \mid j)\; \psi_{j+1}^{g,\theta}\big(h_k'(s;\, j, g, \theta)\big)\, d\mu_j^{g,\theta}(s). \tag{3}$$

Here, $\mu_j^{g,\theta}$ denotes the unconditional population measure across individual states, satisfying $\int d\mu_j^{g,\theta} = \phi_j^g(\theta)$. The index $k \in \{0, 1\}$ indicates whether a health shock occurs, and $h_k'(s;\, j, g, \theta)$ denotes the corresponding next-period health, determined by the individual's state and optimal medical spending, as specified in equation (16). The probability of a health shock is $\Pr(1 \mid j) = \pi_j^h$, as defined in equation (17). The population shares are normalized so that $\sum_g \sum_\theta \sum_{j=1}^J \phi_j^g(\theta) = 1$. Survival beyond the terminal age is zero: $\psi_{J+1}^{g,\theta}(\cdot) \equiv 0$.

Within-period timing. Within each period, events occur in the following order:

1. Households observe their current state (a, h, η).

2. Working-age households choose consumption c, labor supply ℓ, savings a', and medical spending m. Retired households make the same decisions except for labor supply. Medical spending is chosen before the health shock is realized and therefore cannot respond to its occurrence. In this sense, medical spending is preventive rather than curative.

3. End-of-period assets are a', and health after depreciation and medical spending is $\tilde{h} = (1 - \delta_j^h)\, h + H_j(m, h)$, where H_j is the health production function defined in equation (14). The health shock is then realized. With probability π_j^h, the household loses ς units of health, subject to the bounds in equation (16).

Survival to the next period occurs with probability $\psi_{j+1}^{g,\theta}(h')$, which depends on realized next-period health h'. Assets of deceased households are voided rather than transferred to survivors. These assets do not enter aggregate capital in the following period, and their resource implications are captured by Λ_{void} in equation (32).

3.2 Preferences

Individuals derive utility from consumption, leisure, and health. For an individual of sex g and skill type θ entering the economy at age 20, expected discounted lifetime utility is

$$\mathbb{E}_1 \left[\sum_{j=1}^{J} \beta^{j-1} \left(\prod_{i=2}^{j} \psi_i^{g,\theta}(h'_{i-1}) \right) u(c_j,\, \ell_j,\, h_j) \right], \tag{4}$$

where $\beta \in (0,1)$ is the discount factor and $\psi_i^{g,\theta}(h'_{i-1})$ is the probability of surviving from age $i-1$ to age i, conditional on realized health h'_{i-1} at the end of the previous period. The expectation is taken over the paths of idiosyncratic productivity shocks, governed by equation (13), and health shocks, defined in equation (16).

Period utility combines a GHH aggregator over consumption and labor supply with a multiplicative health amenity:

$$u(c,\ell,h) \;\equiv\; \Omega(h) \left[\frac{\left(c - v(\ell)\right)^{1-\gamma} - 1}{1-\gamma} \;+\; \chi \right], \qquad v(\ell) \equiv \Psi \frac{\ell^{1+\nu_\ell}}{1+\nu_\ell}, \tag{5}$$

where the health amenity is given by

$$\Omega(h) \equiv \begin{cases} \exp\!\left(\Xi \dfrac{h^{1-\xi} - 1}{1-\xi} \right) & \xi \neq 1, \\ h^{\Xi} & \xi = 1. \end{cases} \tag{6}$$

Here, $c \geq 0$ denotes consumption, $\ell \in [0,1]$ denotes hours worked, and $1-\ell$ denotes leisure. Health capital is measured as one minus a health deficit, with $h \in (0, \bar{h}]$ and $\bar{h} = 1$ denoting full health. The parameter $\gamma > 0$ governs utility curvature, $\Psi > 0$ determines the scale of labor disutility, and $\nu_\ell > 0$ is the inverse Frisch elasticity. The constant $\chi > -1$ affects the value of being alive, while $\Xi \geq 0$ and ξ govern the responsiveness of utility to health and its curvature, respectively. The health amenity is normalized so that $\Omega(1) = 1$.

The multiplicative health amenity makes health and consumption complements in period utility, since $u_{ch} = \Omega'(h)(c - v(\ell))^{-\gamma} \geq 0$, with strict inequality when $\Xi > 0$. This specification also implies that the marginal rate of substitution between health and consumption, u_h/u_c, increases with consumption when $\Xi > 0$ and the expression in

brackets in equation (5) is nonnegative. Thus, higher consumption can increase households' willingness to pay for improvements in health, providing a channel through which differences in economic resources affect health investment. By contrast, an additive health amenity inside the GHH aggregator would make this marginal rate of substitution independent of consumption. The multiplicative amenity also preserves two properties of GHH preferences: marginal utility of consumption approaches infinity as c approaches $v(\ell)$ from above, and the health amenity cancels from the intratemporal labor-supply condition, preserving the absence of wealth effects on labor supply. Appendix A.1 derives these properties.

The parameter χ affects the value of being alive relative to death, which has zero continuation value. Given the specification in equation (5), the constant component of period utility is $\chi + 1/(\gamma - 1)$, multiplied by $\Omega(h)$. This component influences the value households place on reducing mortality risk and, consequently, their incentives to invest in health. Because χ does not enter the marginal utility of consumption or the intratemporal labor-supply condition, it affects those two decisions through the value of survival. It does enter the marginal utility of health, which is proportional to the bracket in equation (5), so it also raises the value of medical spending directly.

For retirees ($j \geq j_R$), labor supply is zero ($\ell_j = 0$), so period utility simplifies to

$$u(c, 0, h) = \Omega(h) \left[\frac{c^{1-\gamma} - 1}{1 - \gamma} + \chi \right].$$

3.3 Technology

A representative competitive firm produces the economy's single consumption good using capital K and aggregate effective labor L according to the Cobb–Douglas production function:

$$Y = A\, K^{\alpha}\, L^{1-\alpha}, \qquad \alpha \in (0, 1), \tag{7}$$

where $A > 0$ denotes total factor productivity and α is the output elasticity of capital. The production function exhibits constant returns to scale, and capital depreciates at rate $\delta \in (0, 1)$. Aggregate quantities are expressed in stationary per capita terms, as defined in Section 3.6.

Firm's problem. The representative firm takes the interest rate r and the wage per unit of effective labor w as given. It chooses capital K and effective labor L to maximize profits:

$$\max_{K,L} \left\{ A K^{\alpha} L^{1-\alpha} - (r + \delta)\, K - w\, L \right\}. \tag{8}$$

The first-order conditions yield

$$r = \alpha A \left(\frac{K}{L}\right)^{\alpha-1} - \delta, \tag{9}$$

$$w = (1-\alpha) A \left(\frac{K}{L}\right)^{\alpha}. \tag{10}$$

3.4 Mortality, Labor Productivity, and Health

3.4.1 Mortality

Survival probabilities depend on age, sex, education, and health. The probability of surviving from age j to age $j+1$ is given by

$$\psi_{j+1}^{g,\theta}(h') \;=\; \psi_{j+1}^{\circ\, g,\theta} \, \min\Big\{1, \;\; \varpi \;+\; (1-\varpi)\, h'\Big\}, \tag{11}$$

where h' denotes health at the end of period j, $\psi_{j+1}^{\circ\, g,\theta} \in (0,1]$ is the baseline survival probability for individuals of sex g and education θ, and $\varpi \in (0,1)$ determines the fraction of baseline survival retained as health approaches zero. At full health ($h' = 1$), survival equals the baseline probability. Lower health reduces the probability of survival.

Differences in survival across education groups reflect both exogenous differences in baseline survival probabilities and differences in health. The former capture mortality differences that are not generated by medical spending, while the latter reflect endogenous health investment and health shocks.

3.4.2 Labor Productivity

The effective labor productivity of a working-age individual ($j < j_R$) depends on age, sex, education, health, and idiosyncratic productivity:

$$\nu_j(h, \eta;\, g, \theta) \;=\; e_j^{g,\theta} \exp\big(\theta + \eta + \varrho\, h\big), \tag{12}$$

where $e_j^{g,\theta} > 0$ is the sex- and education-specific age-efficiency profile, $\theta \in \{\theta_L, \theta_H\}$ is the education-specific log-productivity intercept, and $\varrho \geq 0$ governs the effect of health on productivity, common across demographic groups. Differences in education affect productivity through both the intercept and the age-efficiency profile, while sex affects productivity through the age-efficiency profile. Productivity is zero for retirees ($j \geq j_R$).

The idiosyncratic component η follows an AR(1) process:

$$\eta_{j+1} = \rho\, \eta_j + \varepsilon_{j+1}, \qquad \varepsilon_{j+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2), \tag{13}$$

where $|\rho| < 1$.

3.4.3 Health Formation, Health Investment, and Health Risk

Health capital depreciates at an age-dependent rate $\delta_j^h \in (0,1)$. Individuals choose medical spending m to replenish their health capital, which is also subject to stochastic health shocks. Depreciation and the probability of health shocks depend on age, while medical spending is an endogenous household decision.

Health production. Medical spending increases health capital according to

$$H_j(m,h) \;=\; \overline{H}_j\, m^{\zeta_h} \underbrace{\frac{\bar{h} - (1-\delta_j^h)\, h}{\bar{h}}}_{\text{remaining deficit}}, \qquad \zeta_h \in (0,1), \tag{14}$$

where $\overline{H}_j > 0$ is an age-dependent productivity parameter and ζ_h governs the diminishing returns to medical spending. The health produced by a given amount of medical spending depends on the health deficit remaining after depreciation. Consequently, medical spending is more productive for individuals with lower health capital. Let $\tilde{h} \equiv (1-\delta_j^h)\, h + H_j(m,h)$ denote health after depreciation and medical spending, but before the realization of the health shock. The effects of current health and medical spending on post-investment health are given by

$$\frac{\partial \tilde{h}}{\partial h} \;=\; (1-\delta_j^h)\left[1 - \frac{\overline{H}_j m^{\zeta_h}}{\bar{h}}\right], \qquad \frac{\partial H_j}{\partial m} \;=\; \zeta_h\, \overline{H}_j\, m^{\zeta_h - 1}\, \frac{\bar{h} - (1-\delta_j^h)h}{\bar{h}}. \tag{15}$$

These derivatives enter the household optimality conditions derived in Appendix A.2.

Stochastic health. In addition to depreciation, individuals face stochastic health shocks. With probability π_j^h, an individual experiences a shock that reduces health capital by $\varsigma > 0$ units. Next-period health is given by

$$h' = \begin{cases} \min\{\tilde{h},\ \bar{h}\} & \text{with probability } 1-\pi_j^h \quad \text{(no shock)}, \\ \min\big\{\max\{\tilde{h}-\varsigma,\ \underline{h}\},\ \bar{h}\big\} & \text{with probability } \pi_j^h \quad \text{(shock)}, \end{cases} \tag{16}$$

where $\underline{h} > 0$ is the minimum level of health following a shock. We denote the two possible realizations of next-period health by h'_k, for $k \in \{0,1\}$, with $\Pr(1 \mid j) = \pi_j^h$. The probability of experiencing a health shock varies with age according to

$$\pi_j^h \;=\; \min\left\{ \bar{\pi}^h\, \frac{\delta_j^h}{\overline{\delta^h}},\ \ \overline{\pi}^{\max} \right\}, \qquad \overline{\delta^h} \equiv \frac{1}{j_R - 1} \sum_{i=1}^{j_R - 1} \delta_i^h, \tag{17}$$

where $\overline{\delta^h}$ is average health depreciation over working ages, $\bar{\pi}^h$ governs the frequency of health shocks, and $\overline{\pi}^{\max}$ is the maximum shock probability. The specification implies that health shocks are more likely at ages with higher depreciation rates. The two estimated parameters governing health risk are $\bar{\pi}^h$ and ς, which determine the frequency and magnitude of health shocks, respectively; the cap $\overline{\pi}^{\max}$ is not estimated.

Conditional on age, all households face the same probability of a health shock, and the shock size ς is common across individuals. Thus, differences in health across households can arise from their medical spending decisions and histories of realized shocks, rather than differences in the probability of experiencing a shock.

Deterministic benchmark. For comparison, setting $\bar{\pi}^h = 0$ eliminates health shocks. If health production is also made independent of the remaining health deficit, the law of motion becomes

$$h' = \min\left\{(1 - \delta_j^h)\, h + \overline{H}_j\, m^{\zeta_h},\ \bar{h}\right\}. \tag{18}$$

In this alternative, the amount of health produced by medical spending does not depend on the individual's current health. The baseline model instead incorporates deficit-scaled health production and stochastic health shocks. Section 4.8 evaluates the quantitative implications of these differences.

3.5 Government

The government operates two separate fiscal accounts. The general government collects taxes, finances public spending, and issues one-period debt. The pay-as-you-go (PAYG) pension system collects payroll contributions from workers to finance pension benefits. Pension contributions do not enter the general government budget but affect households' labor supply decisions through the tax wedge on earnings.

3.5.1 Fiscal Policy

The government levies taxes on consumption, labor income, capital income, and medical spending at rates τ^c, τ^ω, τ^k, and τ^m, respectively. Government spending is proportional to output, $G = g_y Y$, and the government issues one-period debt B. All aggregate quantities are expressed in stationary per capita terms, as defined in Section 3.6. The government's budget constraint is

$$\tau^c C + \tau^m M + \tau^\omega w L + \tau^k r K + B' \;=\; G + (1 + r)B, \tag{19}$$

where C denotes aggregate consumption, M aggregate medical spending, K the aggregate capital stock, and B' government debt in the following period.

Fiscal closure. The tax rates $(\tau^c, \tau^\omega, \tau^k, \tau^m)$ and the government spending share g_y are exogenous policy parameters. In a stationary equilibrium, government debt satisfies $B' = B$, and the government budget constraint becomes

$$r\,B \;=\; \tau^c\,C + \tau^\omega\,w\,L + \tau^k\,r\,K + \tau^m\,M - G. \tag{20}$$

The stock of government debt B adjusts endogenously to satisfy this condition.

3.5.2 Pension System

The pay-as-you-go pension system finances retirement benefits through a payroll contribution rate τ^p on labor income. The pension system balances its revenues and expenditures each period:

$$\tau^p\,w\,L \;=\; \overline{pen}\,N^R, \tag{21}$$

where $\overline{pen}$ is the pension benefit received by each retiree and $N^R = \sum_g \sum_\theta \sum_{j=j_R}^{J} \phi_j^g(\theta)$ is the share of retirees in the population. Pension benefits are determined by

$$\overline{pen} \;=\; \kappa \cdot \frac{wL}{N^W}, \tag{22}$$

where κ is the replacement rate and $N^W = \sum_g \sum_\theta \sum_{j=1}^{j_R-1} \phi_j^g(\theta)$ is the share of working-age individuals in the population. Thus, pension benefits are proportional to average labor income per working-age individual.

This is a stylization of the Mexican system rather than a description of it. Benefits for workers who entered before the reforms of 1997 (private sector) and 2007 (public sector) resemble a pay-as-you-go scheme, but later entrants hold individual accounts. We model a single pay-as-you-go pillar because it isolates the fiscal arithmetic of the dependency ratio, the object this paper prices. Our results therefore read as the cost of aging under a pay-as-you-go benchmark, not as a projection of the legislated system.

Pension closure. The replacement rate κ is an exogenous policy parameter, while the contribution rate τ^p adjusts to balance the pension system each period. Substituting equation (22) into equation (21) yields

$$\tau^p \;=\; \kappa\,\frac{N^R}{N^W}, \tag{23}$$

where N^R/N^W is the pension system's dependency ratio. Thus, for a given replacement rate, an increase in the ratio of retirees to working-age individuals raises the contribution rate, increasing the tax wedge on labor income.

3.6 Markets

Aggregate quantities are obtained by integrating individual decisions over the stationary population distributions $\mu_j^{g,\theta}$. Since the population shares sum to one across ages and demographic types, all aggregates are expressed in stationary per capita terms. Population growth is incorporated into the cohort-weight recursion (2), so aggregate quantities remain constant in a stationary equilibrium even as the total population grows. For example, K denotes capital per person, and the total capital stock at time t is $K_t^{\text{abs}} = KN_t$, where N_t denotes total population. Capital K, household assets A_{dom}, and government debt B are per capita stocks, while output Y, consumption C, medical spending M, government spending G, and effective labor L are per capita flows. The quantities N^W and N^R denote population shares.

Asset markets are incomplete. Households save through a single noncontingent asset and cannot borrow. Thus, next-period asset holdings satisfy

$$a' \geq 0. \tag{24}$$

Factor prices (r, w) are determined jointly with market clearing, subject to the firm's first-order conditions (9)–(10). The following conditions characterize aggregate quantities and market clearing.

Population shares. The shares of working-age individuals and retirees are

$$N^W = \sum_g \sum_\theta \sum_{j=1}^{j_R-1} \phi_j^g(\theta), \qquad N^R = \sum_g \sum_\theta \sum_{j=j_R}^{J} \phi_j^g(\theta). \tag{25}$$

Consumption and medical spending. Aggregate consumption and medical spending are given by

$$C = \sum_g \sum_\theta \sum_{j=1}^{J} \int c_j(s;\, g, \theta)\, d\mu_j^{g,\theta}(s), \tag{26}$$

$$M = \sum_g \sum_\theta \sum_{j=1}^{J} \int m_j(s;\, g, \theta)\, d\mu_j^{g,\theta}(s). \tag{27}$$

Labor market. Aggregate effective labor supply equals the sum of hours worked by working-age individuals, weighted by their labor productivity:

$$L = \sum_g \sum_\theta \sum_{j=1}^{j_R-1} \int \nu_j(h, \eta;\, g, \theta)\, \ell_j(s;\, g, \theta)\, d\mu_j^{g,\theta}(s). \tag{28}$$

Capital market. Aggregate household savings finance productive capital and government debt. Capital-market clearing requires

$$K = A_{\text{dom}} - B, \tag{29}$$

$$A_{\text{dom}} = \sum_{g} \sum_{\theta} \sum_{j=1}^{J} \int a \, d\mu_j^{g,\theta}(s). \tag{30}$$

Here, A_{dom} denotes aggregate household asset holdings, and B is the stock of government debt. This condition, together with the firm's capital-demand condition (9), determines the equilibrium interest rate.

Goods market. Aggregate output is allocated to consumption, medical spending, gross investment, government spending, and resources associated with assets voided at death:

$$Y = C + M + (n_p + \delta) K + G + \Lambda_{\text{void}}, \tag{31}$$

where $(n_p + \delta)K$ is gross investment, accounting for capital depreciation and population growth. The value of assets voided at death is given by

$$\Lambda_{\text{void}} \equiv \sum_{g} \sum_{\theta} \sum_{j=1}^{J} \left(1 - \bar{\psi}_{j+1}^{g}(\theta)\right) \int a_j'(s;\, g, \theta) \, d\mu_j^{g,\theta}(s). \tag{32}$$

As specified in Section 3.1, assets of deceased individuals are not transferred to surviving households.

3.7 Optimization Problems

3.7.1 Net Labor Income

Working-age individuals earn labor income based on their productivity and hours worked. Net labor income is given by

$$y_j(h, \eta, \ell;\, g, \theta) = w \, \nu_j(h, \eta;\, g, \theta) \, \ell \left(1 - \tau^{\omega} - \tau^{p}\right), \tag{33}$$

where w is the wage per unit of effective labor, τ^{ω} is the labor income tax rate, and τ^{p} is the pension contribution rate. Both taxes apply to gross labor earnings. All labor supplied in the model is formal.

3.7.2 Workers' Problem

At each working age $j < j_R$, individuals observe their assets, health, and idiosyncratic productivity and choose consumption, labor supply, savings, and medical spending.

Given labor supply ℓ, resources available at age j are

$$x_j(a, h, \eta, \ell;\, g, \theta) \;=\; \big(1 + r(1-\tau^k)\big)a + y_j(h, \eta, \ell;\, g, \theta). \tag{34}$$

The worker's Bellman equation is

$$\begin{aligned} V_j(a, h, \eta;\, g, \theta) \;=\; \max_{\substack{c\geq 0,\, a'\geq 0 \\ \ell\in[0,1],\, m\geq 0}} \Big\{ & u(c, \ell, h) \\ & + \beta \sum_{k\in\{0,1\}} \Pr(k \mid j)\; \psi_{j+1}^{g,\theta}\big(h'_k\big)\, \mathbb{E}\big[V_{j+1}(a', h'_k, \eta';\, g, \theta) \mid \eta\big] \Big\} \end{aligned} \tag{35}$$

subject to

$$(1+\tau^c)\, c + (1+\tau^m)\, m + a' = x_j(a, h, \eta, \ell;\, g, \theta), \tag{36}$$

$$h'_k \text{ given by (16) with } \tilde{h} = (1-\delta_j^h)\, h + H_j(m, h), \tag{37}$$

$$a' \geq 0, \quad m \geq 0, \quad \ell \in [0, 1]. \tag{38}$$

The continuation value accounts for the two possible realizations of the health shock, with survival probabilities evaluated at the corresponding next-period health levels. The expectation over next-period idiosyncratic productivity is conditional on current productivity, and productivity innovations are independent of health shocks. We denote the optimal savings policy by $a'_j(a, h, \eta;\, g, \theta)$, and analogously for consumption, labor supply, and medical spending.

3.7.3 Retiree's Problem

Retirees ($j \geq j_R$) do not supply labor and receive pension benefits $\overline{pen}$. Their available resources are

$$x_j(a;\, g, \theta) \;=\; \big(1 + r(1-\tau^k)\big)a + \overline{pen}. \tag{39}$$

Retirees choose consumption, savings, and medical spending. The idiosyncratic productivity shock η no longer affects their resources but remains in the state space for consistency across ages. Their Bellman equation is

$$\begin{aligned} V_j(a, h, \eta;\, g, \theta) \;=\; \max_{c\geq 0,\, a'\geq 0,\, m\geq 0} \Big\{ & u(c, 0, h) \\ & + \beta \sum_{k\in\{0,1\}} \Pr(k \mid j)\; \psi_{j+1}^{g,\theta}\big(h'_k\big)\, \mathbb{E}\big[V_{j+1}(a', h'_k, \eta';\, g, \theta) \mid \eta\big] \Big\} \end{aligned} \tag{40}$$

subject to

$$(1+\tau^c)\,c + (1+\tau^m)\,m + a' = x_j(a;\, g, \theta), \tag{41}$$

$$h'_k \text{ given by (16) with } \tilde{h} = (1-\delta_j^h)\,h + H_j(m,h), \tag{42}$$

$$a' \geq 0, \quad m \geq 0. \tag{43}$$

At the terminal age $j = J$, survival to the next period is zero. Consequently, retirees choose $a'_J = 0$ and $m_J = 0$, since neither savings nor medical spending provides any benefit beyond the terminal period.

Optimality. Appendix A.2 derives the household first-order, envelope, and Kuhn–Tucker conditions. Medical spending affects expected lifetime utility through two channels: it increases the probability of survival and raises the value of health conditional on survival, through its effects on future utility, productivity, and survival (equation (53)). Survival probabilities also enter the consumption Euler equation (58), affecting saving incentives. In the absence of insurance against health shocks, households use savings to self-insure against this risk. Finally, the health amenity cancels from the intratemporal consumption–leisure condition (59), so health affects the labor-supply margin directly through productivity rather than through the amenity.

3.8 Stationary Competitive Equilibrium

Let $\mu_j^{g,\theta}$ denote the stationary population measure of age-j individuals of sex g and skill θ over the state space $\mathcal{S} = \mathcal{A} \times \mathcal{H}_h \times \mathcal{H}_\eta$. The measure satisfies $\int d\mu_j^{g,\theta} = \phi_j^g(\theta)$, so aggregate quantities are obtained by integrating directly over the population measures without additional cohort weights. Let $\Pi^\eta(\eta, \eta')$ denote the transition probabilities for idiosyncratic productivity, obtained by discretizing the AR(1) process (13) over the finite state space $\mathcal{H}_\eta$, as described in Appendix B.

3.8.1 Law of Motion for the Distribution

The stationary population distribution evolves according to household decisions, health and productivity shocks, and survival probabilities. For each measurable set $\mathcal{B} \subseteq \mathcal{S}$, age $j = 1, \ldots, J-1$, sex g, and skill type θ, the distribution satisfies

$$\begin{aligned} \mu_{j+1}^{g,\theta}(\mathcal{B}) \;=\; \frac{1}{1+n_p} \int \sum_{k\in\{0,1\}} \Pr(k \mid j)\; \psi_{j+1}^{g,\theta}\big(h'_k(s;j,g,\theta)\big) \\ \Big[\sum_{\eta' \in \mathcal{H}_\eta} \Pi^\eta(\eta,\eta')\, \mathbf{1}\big[\big(a'_j(s;\, g,\theta),\; h'_k(s;j,g,\theta),\; \eta'\big) \in \mathcal{B}\big] \Big]\, d\mu_j^{g,\theta}(s), \end{aligned} \tag{44}$$

where $h'_k(s; j, g, \theta)$ denotes next-period health conditional on the realization of the health shock and optimal medical spending. The indicator function identifies individuals whose next-period state belongs to $\mathcal{B}$. Each individual's contribution to the next-period distribution is weighted by the probability of the corresponding health shock, survival probability, and productivity transition. The factor $1/(1 + n_p)$ accounts for population growth, consistent with the cohort-weight recursion (2).

Individuals enter the economy with zero assets, initial health $h_0 \in (0, \bar{h}]$, and idiosyncratic productivity drawn from the ergodic distribution of Π^η. The initial distribution is

$$\mu_1^{g,\theta}(\mathcal{B}) \;=\; \phi_1^g(\theta) \cdot \sum_{\eta_1 \in \mathcal{H}_\eta} \pi^\eta_{\text{erg}}(\eta_1)\, \mathbf{1}[(0, h_0, \eta_1) \in \mathcal{B}], \tag{45}$$

where π^η_{erg} is the ergodic distribution of idiosyncratic productivity.

3.8.2 Definition

Definition 1 (Stationary Recursive Competitive Equilibrium)**.** *Given exogenous policy* $(\tau^c, \tau^\omega, \tau^k, \tau^m, g_y, \kappa, j_R)$*, birth shares* $\{\pi_1^{g,\theta}\}_{g,\theta}$*, population growth* n_p*, initial health endowment* $h_0 \in (0, \bar{h}]$*, and parameters governing preferences, technology, mortality, productivity, and health formation, a stationary recursive competitive equilibrium consists of*

1. *Value functions* $\{V_j(\cdot; g, \theta)\}_{j=1,\ldots,J;\, g,\theta}$ *and policy functions* $\{c_j, \ell_j, m_j, a'_j\}(\cdot; g, \theta)$ *defined on* $\mathcal{S}$*;*

2. *Factor prices* (r, w)*;*

3. *Government debt* B*, the pension contribution rate* τ^p*, and pension benefits* $\overline{pen}$*;*

4. *Aggregate quantities* $(K, L, C, M, A_{\text{dom}}, \Lambda_{\text{void}})$ *and population shares* (N^W, N^R)*, as defined in Section 3.6;*

5. *Cohort weights* $\{\phi_j^g(\theta)\}_{j,g,\theta}$ *and average survival probabilities* $\{\bar{\psi}_{j+1}^g(\theta)\}_{j,g,\theta}$*;*

6. *Stationary population distributions* $\{\mu_j^{g,\theta}\}_{j,g,\theta}$*;*

such that:

1. Worker optimality. *For each* (g, θ) *and* $j < j_R$*, the value and policy functions solve* (35)–(38).

2. Retiree optimality. *For each* (g, θ) *and* $j \geq j_R$*, the value and policy functions solve* (40)–(43)*, with* $\ell_j \equiv 0$*.*

3. Firm optimality. *Factor prices satisfy* (9)–(10) *given aggregate capital and effective labor.*

4. Government budget. *Government debt satisfies* (20).

5. Pension balance. *The contribution rate and pension benefits satisfy* (23) *and* (22), *respectively.*

6. Market clearing. *The labor, capital, and goods markets clear according to* (28), (29), *and* (31).

7. Cohort weights. *Population shares satisfy* (2), *with average survival probabilities given by* (3) *and entrant shares satisfying* $\phi_1^g(\theta) \propto \pi_1^{g,\theta}$, *normalized so that* $\sum_g \sum_\theta \sum_j \phi_j^g(\theta) = 1$.

8. Stationarity of the distribution. *For each* (g, θ, j), *the population distribution satisfies* (44), *with initial conditions given by* (45).

9. Aggregate consistency. *Aggregate quantities are computed from the stationary population distributions using the definitions in Section 3.6.*

10. Demographic–distributional consistency. *Cohort weights, average survival probabilities, and stationary population distributions jointly satisfy* (2), (3), *and* (44), *evaluated at the equilibrium household policy functions.*

Because survival depends on endogenous health, the stationary age distribution must be consistent with household medical-spending decisions. The resulting ratio of retirees to working-age individuals also determines the equilibrium pension contribution rate through (23).

3.9 Equilibrium Computation

We solve for stationary equilibria numerically. Given model parameters and policy, we solve the household optimization problems by backward induction and compute the associated stationary population distribution, cohort weights, and health-dependent survival probabilities. We then adjust the capital–labor ratio until factor prices and the capital market clear, with the pension contribution rate and government debt determined by their respective budget constraints. Appendix B describes the computational procedure and reports the numerical equilibrium residuals. We do not provide a formal existence proof.

4 Calibration

We calibrate the model to Mexico in two steps. First, we set the demographic and fiscal parameters and specify the baseline survival probabilities, age-efficiency profiles, productivity process, and health depreciation using Mexican data and external evidence. These inputs account for differences across sex, education, and age, as described in the

model. Second, we estimate nine remaining parameters governing preferences, health production, health risk, and the effect of health on labor productivity using classical minimum distance with ten target moments. Section 4.1 describes the data, Section 4.3 the first-step inputs, Section 4.4 the estimation procedure and targets, Section 4.5 the parameter estimates and model fit, and Section 4.6 the model's performance against moments not targeted in estimation.

4.1 Data

We use two complementary Mexican household surveys to calibrate the model. The National Survey of Household Income and Expenditure (ENIGH) provides information on earnings and medical spending, while the Mexican Health and Aging Study (ENASEM) provides information on health and survival at older ages. Table 1 summarizes these surveys and the official and international sources used for the remaining calibration inputs.

The National Survey of Household Income and Expenditure (ENIGH), conducted by INEGI, provides information on household income, expenditures, and demographic characteristics. We use ENIGH to construct the sex- and education-specific age-efficiency profiles, skill intercepts, and population shares of the four demographic types. The survey also provides seven of the ten moments targeted in the second-step estimation: three medical-spending shares, the age gradient of medical spending, two income elasticities of medical spending, and the skill premium. We pool the 2016 and 2018 waves, which overlap with the period covered by the health survey and correspond to the Seguro Popular arrangement represented in the baseline economy. Later waves are excluded from the calibration. All statistics use survey expansion weights, with standard errors clustered at the primary sampling unit level. Household-level moments are assigned to age groups based on the age of the household head.

The Mexican Health and Aging Study (ENASEM, harmonized as MHAS) is a panel survey of adults aged 50 and older. We use the harmonized waves of 2012, 2015, 2018, and 2021 to estimate baseline survival probabilities by age, sex, and education, the survival floor, and age-specific health depreciation. The survey also provides the two survival moments targeted in the second-step estimation. We measure health as one minus an index of health deficits constructed from measures of functional status, depressive symptoms, and self-rated health. These components are selected because they capture aspects of health that can potentially improve with medical care.

We complement the household surveys with three additional data sources. CONAPO life tables provide baseline survival probabilities below age 50, for which education-specific mortality estimates are unavailable. The 2020 INEGI census provides the old-age dependency ratio used to calibrate population growth, while CONAPO population projections provide the 2050 dependency ratio used in the counterfactual experiments. Finally, the World Health Organization's Global Health

Expenditure Database provides Mexico's out-of-pocket share of health spending, which we use to map the model's total cost of medical care to the out-of-pocket expenditures observed in ENIGH. It also provides the publicly financed share of health spending, which is subtracted from government consumption.

ENIGH captures only a fraction of the out-of-pocket medical spending recorded in the national health accounts. We therefore use the survey to characterize the composition of medical spending, its variation over the life cycle, and its relationship with income, rather than to measure aggregate spending levels. Accordingly, the medical-spending moments targeted in estimation consist of expenditure shares, an age gradient, and income elasticities. We do not interpret medical-spending amounts reported in ENIGH as national spending totals.

We obtain the remaining externally assigned parameters from previous studies. The capital depreciation rate and capital share are based on estimates for Mexico. Because ENIGH is a repeated cross-section and does not allow us to estimate individual earnings dynamics, we use estimates of the persistence and innovation variance of idiosyncratic productivity from the existing literature, adjusted to the model's five-year periods. Tax rates are based on effective rates consistent with the tax instruments in the model rather than statutory rates. Section 4.3 describes these parameters and their sources. To evaluate the calibrated equilibrium, we use a capital–output ratio of 3.29 and an investment share of 20.2 percent, both calculated from full-sample averages for Mexico in the Penn World Table, and public pension expenditures of 4.5 percent of GDP, reported by the OECD for 2022.

Table 1: Data sources and calibration inputs

Source	Type	Coverage	Supplies
ENIGH (INEGI)	Household budget survey	Pooled 2016, 2018	Age-efficiency profiles; skill intercepts; birth shares; seven targeted moments
ENASEM/MHAS	Panel, ages 50+	Harmonized 2012, 2015, 2018, 2021	Baseline survival by sex and education; survival floor; health depreciation; two targeted survival levels
CONAPO	Official life tables and projections	Life tables; 2050 projection	Baseline survival below age 50; the 2050 dependency ratio
INEGI Censo	Population census	2020	Old-age dependency ratio, which pins population growth
WHO GHED	Health accounts database	2020	Out-of-pocket share of health spending; public health financing share

Notes. ENIGH provides information on household income and expenditures across age groups, while ENASEM provides longitudinal information on health and survival among older adults. The tenth calibration target, prime-age male hours, is assigned a conventional value rather than estimated from either survey. Externally assigned parameters and aggregate comparison benchmarks are described in Section 4.3.

4.2 Units and Normalization

A model period corresponds to five years. Individuals enter the economy at age 20 ($j = 1$) and retire at age 65 ($j_R = 10$). Output, consumption, medical spending, and labor income are flows measured over five years, while capital and government debt are stocks. To express stock-to-flow ratios at an annual frequency, we multiply the model ratios by five. Thus, the annual capital–output ratio is $5(K/Y)$ and the annual debt–output ratio is $5(B/Y)$. We convert the five-year interest rate to an annual rate using $r^{\text{ann}} = (1 + r)^{1/5} - 1$.

Effective labor is measured in efficiency units, with the age-efficiency of men aged 20–24 normalized to one. The wage w is the price of one efficiency unit of labor. As defined in Section 3.6, aggregate quantities are expressed in stationary per capita terms. We focus on ratios, rates, and comparisons within a given calibration, since absolute levels do not have an independent interpretation across separately normalized calibrations.

4.3 First-Step Inputs

Table 2 summarizes the parameters determined before the second-step estimation, together with their values and sources. These parameters are obtained from Mexican data, external estimates, or model normalizations. We use measures consistent with the model's definitions, including effective rather than statutory tax rates, government consumption net of public health spending, and a population growth rate chosen to match the 2020 old-age dependency ratio. The following paragraphs describe how these inputs are constructed.

Table 2: First-step and externally assigned parameters for Mexico

	Description	Value	Source or calibration basis
Demographics and technology			
	Period length	5 years	Model frequency
$J,\ j_R$	Periods of life; retirement period	17, 10	Maximum age 100; retirement at 65
n_p	Population growth	3.51%/yr (18.8% per 5-yr)	Dependency ratio 65+/(20–64) = 0.1414; INEGI Censo 2020 (see text)
α	Capital share	0.40	Mexico factor shares (see text)
δ	Capital depreciation	5%/yr (22.6% per 5-yr)	Mexico-specific literature value (see text)
A	TFP	1.60	Conditional normalization (see text)
Preferences (fixed externally)			
γ	GHH curvature	2.0	—
ν_ℓ	Inverse Frisch elasticity	2.0	—
β	Discount factor, per 5-yr	0.998^5	External; K/Y-invariant (see text)
Fiscal and pension			
$\tau^c,\ \tau^\omega,\ \tau^k$	Consumption, labor, capital taxes	0.09, 0.10, 0.12	Effective tax rates (see text)
$\tau^m,\ g_y$	Medical tax (baseline); G/Y	0, 0.090	g_y excludes public health spending (see text)
κ	Pension replacement rate	0.40	Flat pension relative to average earnings (see text)
Labor (ENIGH 2016, 2018; this paper)			
$\theta_L,\ \theta_H$	Skill intercepts, log points	−0.020, +0.020	Ages 20–24; earnings-based, divided by $1 + 1/\nu_\ell$
$e_j^{g,\theta}$	Age-efficiency profiles, efficiency units	profiles	Sex- and skill-specific; male 20–24 ≡ 1
$\pi_1^{g,\theta}$	Shares among entrants	shares	Sex-by-education composition at age 20
Productivity and health			
$\rho,\ \sigma_\varepsilon$	AR(1) of η, per 5-yr	0.837, 0.284	External (see text)
δ_j^h	Health depreciation, per 5-yr	$0.0011 \to 0.186$	Age-dependent; ENASEM (see text)
$\psi_j^{\circ\, g,\theta}$	Baseline survival probabilities by sex and education	Age-specific values	ENASEM + CONAPO (see text)
$\bar{\omega}$	Survival floor as $h \downarrow 0$	0.623	ENASEM
h_0	Initial health, health units	$\bar{h} = 1$	—
Measurement			
s	Out-of-pocket share of health spending	0.41	WHO GHED; see Section 4.4

Notes. Annual rates are marked "/yr"; five-year values correspond to the model period. Tax rates are effective rather than statutory rates. The pension contribution rate τ^p is determined endogenously and is therefore not listed. The nine parameters estimated in the second step are reported in Table 4. The range of δ_j^h covers ages 20–24 through the terminal age groups, with depreciation constant from age 90 onward.

Technology and demographic scale. We set capital depreciation at 5% annually, following Bergoeing et al. (2002), who validate this value against the UN-reported share

of capital consumption in Mexican GDP and reject an alternative rate of 8%. The capital share is $\alpha = 0.40$, following García-Verdú (2005). We normalize total factor productivity to $A = 1.60$. Because health is bounded and period utility includes a flow value of being alive, the model is non-homothetic, so changes in the scale of output are not neutral. The estimated parameters are therefore conditional on this normalization. We choose population growth to reproduce the 2020 census old-age dependency ratio. An annual growth rate of $n_p = 3.51\%$ yields a dependency ratio of 0.1414 under the model's survival probabilities. This growth rate exceeds Mexico's historical population growth over the decades that shaped its 2020 age distribution, reflecting the use of a stationary population to represent that distribution. We recompute n_p by bisection whenever survival probabilities change, so the dependency ratio is matched by construction and is not used for validation in Section 4.6. Finally, we fix the five-year discount factor at $\beta = 0.998^5$ rather than calibrating it to the capital–output ratio, which is an equilibrium outcome while the second-step estimation holds factor prices fixed. The calibrated capital stock is relatively insensitive to β, as the equilibrium interest rate adjusts and survival probabilities substantially affect effective discounting at ages when households hold assets.

Mortality. We estimate baseline survival probabilities $\psi_j^{\circ\, g,\theta}$ separately by sex and education using ENASEM. For individuals aged 50 and older, we estimate survival probabilities from deaths observed between survey waves. To account for differences by education, we include an education indicator and its interaction with age, which helps obtain estimates for older age groups with relatively few highly educated individuals. For ages below 50, we use CONAPO life tables without distinguishing education groups, as education-specific mortality estimates are unavailable for these ages. Consequently, education differences in baseline survival probabilities enter the model only from age 50 onward. We also use ENASEM to estimate the survival parameter $\varpi = 0.623$ from the relationship between mortality and the observed health-deficit index. As defined in equation (11), this parameter determines the fraction of baseline survival retained as health approaches zero.

Labor productivity. We use the pooled ENIGH 2016 and 2018 cross-sections to estimate age-efficiency profiles by sex and education, together with skill-specific productivity intercepts. We relate log earnings to the components of individual productivity in equation (12) using

$$\log w_{i,t} = \underbrace{\log e_{j(i,t)}^{g,\theta}}_{\text{age-by-sex-by-skill profile}} + \underbrace{\theta_i}_{\text{skill intercept}} + \underbrace{\varrho\, h_{i,t}}_{\text{health–productivity slope}} + \eta_{i,t}, \tag{46}$$

where the age-efficiency profile is normalized to one for men aged 20–24. The estimated profiles differ considerably across education groups. For low-skilled men, productivity

peaks at ages 45–49 and returns to approximately its initial level by ages 60–64. For high-skilled men, productivity roughly doubles by its peak in the fifties. These profiles incorporate the observed life-cycle evolution of the skill premium into the model. Because we estimate the profiles from earnings rather than hourly wages, we account for the labor-supply response implied by GHH preferences. Under this specification, productivity differences translate into earnings differences through the factor $1+1/\nu_\ell = 1.5$. We therefore divide the estimated log-earnings differences by this factor before incorporating them into the model. An estimation based on hourly wages would avoid this adjustment but is not implemented here.

ENIGH is a repeated cross-section and therefore does not allow us to estimate the persistence and innovation variance of idiosyncratic productivity. We instead use estimates from the restricted-income-profile literature (Storesletten et al., 2004; Guvenen, 2009; Hubbard et al., 1994). Averaging representative estimates gives an annual persistence of $\rho_{\text{ann}} \approx 0.965$ and an annual innovation variance of $\sigma^2_{\varepsilon,\text{ann}} \approx 0.0185$. We convert these estimates to the model's five-year frequency using skip-sampling, obtaining $(\rho, \sigma_\varepsilon) = (0.837, 0.284)$, with the innovation scale adjusted so that the stationary dispersion of log earnings matches that observed among Mexican formal-sector workers (Puggioni et al., 2022). The calibrated capital stock is particularly sensitive to these parameters, as the persistence of idiosyncratic income risk is its dominant determinant.

We estimate the health–productivity parameter ϱ jointly with the other second-step parameters rather than obtaining it from an external wage regression. ENASEM and comparable Latin American health surveys, including Costa Rica's CRELES and Panama's SABE study, begin at age 50 and therefore do not provide health information over most of the working life, unlike the US panel data used by Hosseini et al. (2022). Moreover, health is endogenous to medical spending and consequently correlated with persistent productivity shocks, making an external wage regression unsuitable for identifying ϱ. Within the structural model, the parameter is identified jointly with the health scale, as discussed in Section 4.4.

Health depreciation. Individuals enter the economy at full health, $h_0 = \bar{h} = 1$. We construct the age-specific health depreciation rates δ^h_j from changes in the health-deficit index among ENASEM respondents observed in consecutive survey waves. Because these changes are measured among survivors, the resulting profile is interpreted as a lower bound on health deterioration, since individuals experiencing larger health declines are less likely to survive between waves. We convert the observed three-year changes to the model's five-year frequency. For younger ages not covered by ENASEM, we extrapolate the depreciation rates using a log-linear age profile. The resulting five-year depreciation rate increases from 0.0011 at ages 20–24 to 0.186 at age 90 and remains constant thereafter.

Fiscal block. We set the tax parameters using effective rates that correspond to the tax instruments in the model. The consumption tax is $\tau^c = 0.09$, based on Mexican VAT and excise revenues relative to private consumption. The OECD reports a VAT revenue ratio of approximately 0.35 for Mexico, indicating that the statutory rate of 16% substantially overstates the effective tax burden. We set the labor income tax at $\tau^\omega = 0.10$, close to the OECD's income-tax-only wedge of 9.4% in 2020 and effective estimates of 7.7–8.5% excluding social-security contributions (Bachas et al., 2022). Pension contributions are determined separately through equation (23) and equal 5.66% in the baseline equilibrium. The resulting combined labor tax wedge is approximately 15.7%, compared with a measured total near 20% that includes employer contributions outside the model. Because the labor income tax is flat, the model does not capture distributional effects arising from income-tax progressivity. The capital income tax is $\tau^k = 0.12$, consistent with effective estimates for Mexico (0.108–0.121; Bachas et al., 2022) and the marginal rates reported by Antón Sarabia (2005).

We set the government consumption share at $g_y = 0.090$, calculated as government consumption of 12.2% of GDP in 2020 less publicly financed health spending of 3.2% of GDP. This adjustment separates general government consumption from the medical expenditures represented in the model, as discussed in Section 4.4. Finally, we set the pension replacement rate at $\kappa = 0.40$, following Bonthuis (2024). In the model, this parameter determines a flat pension benefit relative to average labor income per working-age individual. Government debt B and the pension contribution rate τ^p are determined endogenously through equations (20) and (23), respectively.

4.4 Second-Step Estimation

We estimate nine parameters jointly using classical minimum distance. The estimated parameter vector is

$$\Theta_{\text{MD}} = \big(\Psi,\ \Xi,\ \overline{H}_0,\ h^{\text{slope}},\ \zeta_h,\ \varrho,\ \chi,\ \bar{\pi}^h,\ \varsigma\big).$$

These parameters govern labor disutility (Ψ), the health amenity (Ξ), health production ($\overline{H}_0$, h^{slope}, ζ_h), the effect of health on labor productivity (ϱ), the flow value of being alive (χ), and the frequency and magnitude of health shocks ($\bar{\pi}^h$, ς). The age profile of health-production productivity is specified as

$$\overline{H}_j = \overline{H}_0 \exp\big(-h^{\text{slope}}\, j\big), \qquad \zeta_h \in (0,1), \tag{47}$$

where h^{slope} determines how health-production productivity varies with age, with its sign estimated rather than imposed. We hold the health-amenity curvature parameter ξ fixed at an interior value and set the discount factor β externally, rather than estimating either parameter in this step.

We estimate the parameters by minimizing the sum of squared proportional devia-

tions between the ten model moments and their empirical counterparts. This criterion assigns equal weight to equally sized percentage deviations across targets. Appendix B presents the objective function, explains the choice of proportional rather than inverse-variance weighting, and describes the numerical optimization procedure.

Targets. Table 3 reports the ten moments targeted in estimation. Nine are constructed from Mexican survey data, while the target for prime-age male hours is assigned a conventional value. The hours target disciplines the scale of labor disutility. Three medical-spending shares for ages 25–34, 50–64, and 65–79, together with the age gradient of medical spending, discipline the level and age profile of medical expenditures. The age gradient constrains the shape of the profile, while the expenditure shares inform its level across age groups. Two income elasticities discipline the response of medical spending to income, which is central to the model's endogenous health-investment mechanism. The skill premium at ages 25–34 helps discipline the effect of health on earnings. Finally, survival probabilities for low- and high-education individuals aged 50–64 discipline mortality outcomes, although these targets overlap with the data used to construct the first-step survival probabilities.

Table 3: Calibration targets and data sources

Targeted Moment	Target	SE	Source
Labor			
Prime-age male hours, share of time endowment	0.333	—	convention (see text)
Medical spending			
Income elasticity of medical spending, pooled	0.575	0.029	ENIGH 2016–2018
Income elasticity of medical spending, within age	0.573	0.029	ENIGH 2016–2018
Medical-spending share of expenditure, ages 25–34	0.0182	0.0004	ENIGH 2016–2018
Medical-spending share of expenditure, ages 50–64	0.0231	0.0004	ENIGH 2016–2018
Medical-spending share of expenditure, ages 65–79	0.0332	0.0009	ENIGH 2016–2018
Log-slope of mean medical spending, ages 25–79	0.0675	0.0082	ENIGH 2016–2018
Earnings			
Skill premium in log earnings, ages 25–34	0.659	0.013	ENIGH 2016–2018
Survival			
Five-year survival, low education, ages 50–64	0.951	0.004	ENASEM
Five-year survival, high education, ages 50–64	0.972	0.006	ENASEM

Notes. Standard errors are clustered at the primary sampling unit level. The standard error for the medical-spending age gradient is obtained by clustered bootstrap, while those for the survival moments use the delta method applied to clustered estimates of survival between survey waves. The prime-age male hours target corresponds to a 40-hour workweek, or one-third of the time endowment. It is assigned rather than estimated from survey data and therefore has no standard error. The ten moments jointly discipline the nine estimated parameters; there is no one-to-one correspondence between individual moments and parameters.

Measurement. ENIGH records out-of-pocket medical spending, whereas m in the model represents the total resource cost of medical care. To make the two measures comparable, we multiply model medical spending by $s = 0.41$, the out-of-pocket share of total health expenditure in Mexico. This measurement adjustment does not affect the household budget constraint, in which medical spending enters as $(1 + \tau^m)m$. Publicly financed health spending is accounted for separately by excluding it from government consumption, as described in Section 4.3.

We estimate the two income elasticities of medical spending using Poisson pseudo-maximum likelihood (PPML), regressing medical spending in levels on log labor income and wave fixed effects. The sample includes working-age household heads with positive labor income, retaining households with zero medical spending. Unlike a log-log regression restricted to positive spending, which would exclude nearly half the sample, PPML estimates the elasticity of conditional mean spending while retaining zeros. Labor income corresponds to $wv\ell$ in the model. For the within-age elasticity, we estimate the regression separately by age group and average the coefficients using population weights.

The medical-spending shares are expansion-weighted averages of medical spending relative to total monetary expenditures. Their model counterpart is $(1 + \tau^m)m/[(1 + \tau^c)c + (1 + \tau^m)m]$. The skill premium is the difference in log labor earnings between high- and low-education individuals aged 25–34. We construct the corresponding model moment using the same regression specification as in the data.[1]

We calculate the survival moments from deaths observed between ENASEM waves among individuals aged 50–64, separately by education, and convert the estimated probabilities to the model's five-year frequency. The model's survival probabilities are anchored using the survival adjustment evaluated at the mean observed health-deficit index. This adjustment enters through the first-step mortality inputs rather than through a direct comparison between observed and model health levels.

We do not target the cross-sectional dispersion or persistence of medical spending. Although we construct both measures from the data, they are not directly comparable to their model counterparts because ENIGH records out-of-pocket expenditures over three months, whereas the model describes medical spending over five years. Section 4.6 discusses this limitation in greater detail. Consequently, the frequency and magnitude of health shocks have no dedicated calibration moments and are jointly disciplined by the medical-spending shares and survival probabilities.

Identification. The ten calibration moments jointly discipline the nine estimated parameters, with no one-to-one correspondence between individual moments and parameters. The prime-age male hours target disciplines the scale of labor disutility, Ψ.

[1] Both earnings moments condition on positive earnings in the data, while the model has no participation margin and averages over all states. Participation differs by education and sex, and the resulting selection is not corrected.

The medical-spending shares, age gradient, and two income elasticities jointly discipline the health-production parameters $(\overline{H}_0, h^{\text{slope}}, \zeta_h)$ and, through their effects on medical spending, the health-amenity parameter Ξ and the flow value of being alive, χ. In particular, χ has no separate calibration target and is disciplined only indirectly through medical-spending decisions. The skill premium helps discipline the health-productivity parameter ϱ, jointly with the education differences in health generated by the model. Finally, the survival probabilities and medical-spending profile jointly discipline the frequency and magnitude of health shocks, governed by $\bar{\pi}^h$ and ς. The estimation procedure also includes a smooth penalty that excludes parameter values under which any household would assign a negative value to survival.

The health-productivity parameter ϱ is estimated jointly with the model-implied differences in health across education groups. Because health is measured in model-specific units rather than directly matched to an observed health measure, the estimate of ϱ is conditional on the model's health normalization and cannot be directly compared with estimates from regressions of earnings on observed health.

Estimation in partial equilibrium. We estimate the parameters by solving the household optimization problems at fixed factor prices, without imposing market clearing for each candidate parameter vector. After estimation, we hold the estimated parameters fixed and solve for the stationary general equilibrium, allowing factor prices and the pension contribution rate to adjust endogenously. We also recalibrate population growth to match the 2020 old-age dependency ratio under the resulting survival probabilities. The calibration and validation results reported in the following subsections are evaluated at these equilibrium prices, which differ from those used in estimation. Evaluating the estimated parameters at general-equilibrium prices improves the minimum-distance objective relative to the fixed-price estimation. The counterfactual experiments in Section 5 are also solved in general equilibrium.

4.5 Calibration Results

Table 4 reports the nine estimated parameters, and Table 5 compares the model moments with their empirical targets. We evaluate the model moments at the baseline general-equilibrium prices, with population growth adjusted to match the 2020 old-age dependency ratio. We do not report standard errors for the parameter estimates for the reasons discussed in Appendix B. Two parameter estimates are treated as boundary solutions and are discussed separately below.

Table 4: Second-step parameter estimates

	Description	Value
Preferences and labor		
Ψ	Labor-disutility scale	24.5
Ξ	Health-amenity scale	$0.0017^{\dagger}$
χ	Flow value of being alive	29.2
Health technology		
$\overline{H}_0$	Health-production scale	3.36
h^{slope}	Age slope of health production	−0.0188
ζ_h	Health-production curvature	0.489
Health risk		
$\bar{\pi}^h$	Health-shock arrival parameter	$0.0100^{\dagger}$
ς	Shock size	0.283
Earnings		
ϱ	Health–productivity slope	0.983
ξ	Amenity curvature	0.50 (pinned)

Notes. The table reports the nine parameters estimated by minimum distance. The amenity curvature ξ is fixed rather than estimated. Parameters marked with † are discussed as boundary solutions in the text. Standard errors are not reported for the reasons discussed in Appendix B.

The interior estimates. The estimated health-production scale is $\overline{H}_0 = 3.36$, and the curvature parameter $\zeta_h = 0.489$ implies diminishing returns to medical spending. The estimated age slope is $\widehat{h}^{\text{slope}} = -0.0188$, implying that the productivity of medical spending increases by approximately 1.9% per five-year period, or roughly one-third over the life cycle. This increase partially offsets the rise in health depreciation with age. The estimated flow value of being alive is $\widehat{\chi} = 29.2$, which contributes to the incentives for medical spending through the value of survival. The estimated health-shock magnitude is $\widehat{\varsigma} = 0.283$, although its interpretation is conditional on a shock-arrival rate estimated at its lower bound. Finally, the health-productivity parameter is $\widehat{\varrho} = 0.983$, indicating a substantial relationship between health and labor productivity in the calibrated model.

The estimated health-amenity scale, $\widehat{\Xi} = 0.0017$, is close to zero, while the health-shock arrival parameter, $\widehat{\bar{\pi}}^h = 0.0100$, reaches its lower bound. These estimates should be interpreted cautiously, particularly because the calibration does not target the dispersion or persistence of medical spending, which could provide additional information about health risk.

Table 5: Fit of targeted calibration moments

Moment	Data	Model	Model/Data
Labor			
Prime-age male hours (convention)	0.333	0.326	0.98
Medical spending			
Income elasticity of m, pooled	0.575	0.724	1.26
Income elasticity of m, within age	0.573	0.831	1.45
Spending share, ages 25–34	0.0182	0.0211	1.16
Spending share, ages 50–64	0.0231	0.0225	0.97
Spending share, ages 65–79	0.0332	0.0203	0.61
Log-slope of mean m, ages 25–79	0.0675	0.0885	1.31
Earnings			
Skill premium, ages 25–34	0.659	0.717	1.09
Survival			
Five-year survival, low education, 50–64	0.951	0.995	1.05
Five-year survival, high education, 50–64	0.972	0.998	1.03

Notes. Model moments are computed from the stationary population distribution at $\widehat{\Theta}_{\text{MD}}$, evaluated at the baseline general-equilibrium prices and population growth calibrated to the 2020 dependency ratio. Although the model closely reproduces the two survival levels in proportional terms, their difference is only about one seventh of the observed education gap in survival. Section 4.6 examines this discrepancy.

Fit of targeted moments. The model reproduces five of the ten calibration moments within 10% of their targets: prime-age male hours, the medical-spending share at ages 50–64, the skill premium, and the two survival probabilities. The largest discrepancies concern medical spending. The model overpredicts the pooled and within-age income elasticities by 26% and 45%, respectively, and the age gradient of medical spending by 31%. It also overpredicts the medical-spending share at ages 25–34 by 16%, while underpredicting the share at ages 65–79 by 39%. Under the proportional minimum-distance objective, the within-age income elasticity and the medical-spending share at ages 65–79 together account for nearly two-thirds of the residual, whereas the two survival probabilities account for less than 1%.

The model does not reproduce the observed increase in medical spending as a share of household expenditures with age. Although medical spending rises with age in response to higher health depreciation, consumption increases more rapidly, leaving the medical-spending share nearly flat. This discrepancy, together with the model's excessive responsiveness of medical spending to income, highlights the limitations of the calibrated health-investment mechanism.

4.6 Validation

We evaluate the calibrated model against additional empirical outcomes reported in Table 6. The value of a statistical life is not targeted in estimation, while the education gap in survival is not directly targeted but depends on survival probabilities that enter the calibration. We also compare the model's male–female earnings gaps with the data, although these differences are partly determined by the age-efficiency profiles calibrated in the first step. Finally, we do not compare the model's health capital directly with the observed health index because the two measures are not expressed on a common scale.

Table 6: Model validation against empirical outcomes

	Model	Data
Panel A. Not directly targeted		
Value of a statistical life, multiple of output per capita	179	152
Education gap in five-year survival (H−L), ages 50–64	+0.003	+0.021
Panel B. Carried from the first step		
M−F log earnings gap, ages 25–34	+0.393	+0.290
M−F log earnings gap, ages 35–49	+0.420	+0.382
M−F log earnings gap, ages 50–64	+0.370	+0.362

Notes. Panel A reports outcomes not directly targeted in estimation. The value of a statistical life is compared with the benefit-transfer estimate of Viscusi and Masterman (2017). The education gap in survival is calculated from ENASEM and is not directly targeted, although the two survival probabilities used to construct it enter the estimation objective. Panel B compares male–female log earnings gaps from ENIGH 2016–2018, calculated among earners using expansion weights, with their model counterparts. These comparisons are not independent validation exercises because the first-step calibration incorporates the observed earnings profiles. Model values are evaluated at the baseline general-equilibrium prices.

Value of a statistical life. We calculate the model-implied value of a statistical life as the mass-weighted trimmed mean, over the middle fifth of the population distribution, of the marginal rate of substitution between mortality risk and money. Expressed as a multiple of annual output per capita, the model yields a value of 179, compared with an external benchmark of 152 from Viscusi and Masterman (2017). The model's estimate is therefore approximately 18% higher than the benchmark. The value of a statistical life is not targeted in estimation, although the parameter χ, which affects the value of survival, is indirectly disciplined by the medical-spending moments. The estimation procedure rules out negative valuations of survival but does not restrict their magnitude.

The education gap in survival. The model generates an education gap in five-year survival of 0.30 percentage points at ages 50–64, compared with 2.09 percentage points

in the data. Of the model-implied gap, 0.24 percentage points are attributable to differences in baseline survival probabilities across education groups, while only 0.06 percentage points arise from endogenous health investment. This decomposition holds policies and the population distribution fixed and replaces education-specific baseline survival probabilities with pooled probabilities. Part of the discrepancy reflects the construction of the baseline survival probabilities. Capping these probabilities at one eliminates the estimated education gradient for women aged 50–69 and reduces it for men. In addition, health capital remains close to its upper bound, limiting the differences in survival generated by endogenous health investment. Despite these limitations, the model predicts that high-education individuals live longer than their low-education counterparts, with differences in expected age at death of 2.5 years for men and 1.9 years for women. These differences in longevity are relevant for the distributional effects of pension reform examined in Section 5. Section 4.8 examines the limitations of the endogenous health mechanism in greater detail.

The gender structure of earnings. Panel B of Table 6 compares the model-implied male–female earnings gaps with those observed in the data. These comparisons are not independent validation exercises because gender differences enter the model through the sex-specific age-efficiency profiles, baseline survival probabilities, and entrant shares calibrated in the first step. All structural parameters are common across sexes. When constructing the productivity inputs, we divide estimated log-earnings differences by $1 + 1/\nu_\ell = 1.5$ to account for the labor-supply response implied by GHH preferences. The model reproduces the observed pattern of an earnings gap that widens between ages 25–34 and 35–49 before narrowing at ages 50–64. However, it overpredicts the magnitude of the gap in all three age groups, particularly among younger individuals. This discrepancy most plausibly reflects selection among earners in the construction of the female productivity profiles. These differences in earnings are relevant for the distributional effects of pension reforms examined in Section 5.

Second moments of spending. We do not evaluate the model against the cross-sectional dispersion or persistence of medical spending. ENIGH records out-of-pocket medical expenditures over a three-month period, whereas each model period spans five years. Moreover, most surveyed households report no medical spending during the reference period, while the model generally implies positive medical spending before the terminal age. These differences make the corresponding moments difficult to compare directly without additional assumptions about short-term medical expenditures or measurement error.

4.7 The Calibrated Baseline Equilibrium

Table 7 reports the stationary general equilibrium of the calibrated model for Mexico in 2020. The equilibrium is computed using the estimated parameters $\widehat{\Theta}_{\text{MD}}$ and the first-step inputs. The model moments reported in Sections 4.5–4.6 are evaluated at these equilibrium prices. This equilibrium also serves as the baseline for the counterfactual experiments in Section 5.

Table 7: Baseline stationary equilibrium for Mexico, 2020

Quantity	Value
Stationary aggregates and prices	
K (capital per capita)	5.16
L (effective labor)	6.51
Y (output)	9.49
r (annual)	8.58%
w (wage per efficiency unit)	0.875
τ^p (PAYG contribution rate)	5.66%
$\overline{pen}$ (pension per retiree)	0.475
C/Y	61.1%
M/Y	3.21%
N^R/N^W (PAYG dependency ratio)	0.141
$5K/Y$ (annual capital–output ratio)	2.72
$5B/Y$ (annual debt–output ratio)	112%

Notes. The table reports the stationary general equilibrium evaluated at the estimated parameters in Table 4, using the estimation grid. Aggregate quantities are expressed in per capita terms. The pension contribution rate τ^p and government debt B are determined endogenously by their respective budget constraints. Capital–output and debt–output ratios are expressed at an annual frequency. Lifetime utility at birth is not reported because welfare comparisons are based on changes in utility across counterfactual equilibria (Section 5.2).

The baseline equilibrium reproduces the 2020 old-age dependency ratio of 0.1414 by construction, since population growth is calibrated to match this target. Other aggregate outcomes differ from their empirical benchmarks. The annual capital–output ratio is 2.72, compared with 3.29 in the data, a shortfall of approximately 17%. The investment share is 22.6% of output, compared with 20.2% in the data, partly reflecting the high population growth rate and longevity in the calibrated economy. Finally, government debt amounts to 112% of annual output, substantially above Mexico's federal debt ratio of approximately 50%. This discrepancy reflects the model's fiscal closure, in which government debt adjusts to satisfy the stationary government budget constraint rather than being targeted in calibration.

4.8 Performance of the Health Block

Under the deterministic health-accumulation benchmark in equation (18), more than 99% of working-age individuals accumulate health at its upper bound, while the medical-spending share among young households falls to approximately 3% of its empirical counterpart. Introducing deficit-scaled health production and stochastic health shocks reduces the fraction of working-age individuals at the health ceiling to approximately 75%. These features generate greater heterogeneity in health across households, although the estimated health distribution remains concentrated near full health.

Despite these improvements, health remains close to its upper bound in the calibrated equilibrium. Across the four demographic types and all seventeen age groups, health levels range from 0.990 to 1.000. At ages 60–64, for example, health is 0.9970 for high-skilled men and 0.9963 for low-skilled men. These small differences limit the contribution of endogenous health investment to differences in survival across education groups, as documented in Section 4.6. In the calibrated model, medical spending primarily offsets health depreciation rather than generating substantial differences in health across individuals. Appendix D documents the predominance of curative spending in ENIGH 2024, which is relevant when interpreting the model's assumption that medical spending is chosen before the current-period health shock.

5 Experiments

Mexico's old-age dependency ratio is projected to more than double, from 0.141 in 2020 to 0.307 in 2050 (CONAPO, 2023). Under the PAYG financing rule (23), maintaining the replacement rate at $\kappa = 0.40$ would increase the pension contribution rate from 5.66% to approximately 12.3%. We use the calibrated model to evaluate the macroeconomic and welfare consequences of demographic aging under this adjustment and compare them with two pension reforms: reducing the replacement rate and raising the retirement age.

We conduct three general-equilibrium experiments using the calibrated 2020 economy as the baseline. All three consider the projected demographic conditions for 2050 but differ in how the pension system adjusts. Under the contribution-rate adjustment (CF0), the replacement rate and retirement age remain unchanged, while the contribution rate increases to balance the pension system. Under the benefit reduction (CF1), the replacement rate is reduced to restore the contribution rate to its 2020 level. Under the retirement-age increase (CF2), the retirement age rises from 65 to 70, while the replacement rate remains unchanged and the contribution rate adjusts endogenously. These experiments allow us to compare the macroeconomic and welfare consequences of adjusting pension contributions, benefits, and retirement eligibility in response to demographic aging.

Each counterfactual is solved as a stationary general equilibrium using the calibrated model. We hold the estimated parameters $\widehat{\Theta}_{\text{MD}}$ and the externally assigned parameters fixed, except for the demographic and pension-policy parameters that change in each experiment. Factor prices, the pension contribution rate, pension benefits, and government debt are determined endogenously. The counterfactual solver reproduces the calibrated 2020 equilibrium when evaluated at the baseline inputs. All comparisons are between stationary equilibria rather than transitions between them, and all equilibria are computed using the estimation grid described in Appendix B.

5.1 Design

The aged economy. We introduce population aging by changing the population growth rate n_p, which determines the stationary age distribution together with survival probabilities. In the baseline, we choose n_p to reproduce the 2020 census old-age dependency ratio of 0.1414. For the 2050 experiments, we choose it to target CONAPO's projected dependency ratio of 0.3065, lowering the implied annual population growth rate from 3.51% to 1.48%. This adjustment allows us to match the projected dependency ratio in a stationary economy, rather than representing a forecast of Mexico's fertility or population growth. Appendix C describes the demographic inputs and their implementation.

The demographic change in all three counterfactuals operates exclusively through the population growth rate n_p. Reducing this parameter changes the stationary age distribution through the cohort-weight recursion (2), increasing the relative size of older cohorts. We hold the baseline survival probabilities $\psi_j^{\circ g,\theta}$ and the composition of entering cohorts $\pi_1^{g,\theta}$ at their 2020 values. This isolates the effects of population aging induced by lower population growth, without incorporating projected changes in mortality or educational composition. Survival nevertheless remains endogenous because medical spending affects health and, consequently, survival probabilities through equation (11).

The two reforms. Both reforms use the demographic assumptions of the aged economy and are evaluated relative to the contribution-rate adjustment (CF0). Under the benefit reduction (CF1), we lower the replacement rate κ to the value that restores the pension contribution rate to its 2020 level of 5.658%, as determined by equation (23). Under the retirement-age increase (CF2), we raise the retirement period from $j_R = 10$ to $j_R = 11$, extending working age from 65 to 70. The replacement rate remains at $\kappa = 0.40$, and the contribution rate adjusts endogenously.

Implementing CF2 requires extending the age-efficiency profiles to ages 65–69, since the baseline profiles end at ages 60–64. We estimate the extension separately for each sex and education group using ENIGH earnings regressions for employed individuals aged 20–69. Specifically, we apply the estimated ratio of age-efficiency at ages 65–69 to

that at ages 60–64 to the corresponding baseline profile, leaving productivity at earlier ages unchanged. The estimated ratios range from 0.87 to 0.96, implying a decline in productivity of 4–13%. These estimates require caution because employment at ages 65–69 is only 19.2%, compared with 28.5% at ages 60–64. The observed earnings profiles therefore reflect selection into employment and may not represent the productivity of individuals who would work under the reform.

The three experiments examine distinct pension-policy adjustments, changing one policy margin at a time. We do not consider financing pensions through general tax revenues, changing the benefit formula, expanding contribution coverage through formalization, or combining reforms. General-revenue financing would require a different fiscal arrangement, while changes to the benefit formula or contribution coverage would extend the current pension structure (Joubert, 2015). We study the three adjustments separately because their general-equilibrium effects cannot be inferred from the PAYG financing identity alone.

The contribution-rate adjustment (CF0) and benefit reduction (CF1) affect households through changes in factor prices, pension contributions, and pension benefits. Neither population growth n_p nor the replacement rate κ enters the household optimization problem directly when these prices and fiscal variables are held fixed. By contrast, the retirement-age increase (CF2) also changes the household problem directly by extending the ages at which individuals can supply labor and delaying pension eligibility.

5.2 The Welfare Metric

We measure welfare differences across stationary equilibria using consumption equivalents, following De Nardi et al. (2017). Consider two economies A and B, and suppose that a newborn of type (g, θ) has higher expected lifetime utility in economy A. We define $\lambda_{g,\theta}$ as the constant fraction of consumption the newborn would give up at every age and in every state in economy A, holding the equilibrium allocation of labor and health fixed, to attain the same expected lifetime utility as in economy B:

$$\mathbb{E}_1\left[\sum_{j=1}^{J}\beta^{j-1}\left(\prod_{i=2}^{j}\psi_i^{g,\theta}(h'_{i-1})\right)u\left((1-\lambda_{g,\theta})\,c_j^A,\,\ell_j^A,\,h_j^A\right)\right] \;=\; \mathcal{W}_1^B(g,\theta), \tag{48}$$

where (c^A, ℓ^A, h^A) denotes the equilibrium allocation in economy A. We use this compensating measure because, with $\gamma > 1$, period utility is bounded above in consumption and differences in survival cannot necessarily be offset by increasing consumption in the worse economy. We report consumption equivalents for each demographic type and aggregate them using the entrant shares $\pi_1^{g,\theta}$. The reported values are signed: $\lambda > 0$ indicates that the first-named economy provides higher welfare, while $\lambda < 0$ indicates the opposite.

We also calculate age-specific consumption equivalents, λ_j, using expected remaining lifetime utility. For each age group j, we determine the fraction of remaining lifetime consumption that residents of the higher-welfare economy would give up to attain the same average remaining lifetime utility as residents of the same age in the comparison economy. These calculations integrate over the stationary distribution of individual states at each age. Importantly, these are comparisons between residents of different stationary equilibria, not welfare effects experienced by cohorts alive when a reform is implemented, since the model does not compute transition paths. At advanced ages, the shorter remaining horizon and the limited ability of consumption to compensate for differences in survival can generate large consumption-equivalent welfare measures. These estimates should therefore be interpreted with caution.

5.3 Cost of Contribution-Rate Adjustment

Table 8 compares the calibrated 2020 equilibrium with the three counterfactual stationary equilibria under 2050 demographics. We begin with the contribution-rate adjustment (CF0).

Table 8: Baseline and counterfactual stationary equilibria

	Baseline	CF0	CF1	CF2
	2020	Contribution-rate adjustment	Benefit reduction	Retirement at 70
Imposed parameters				
n_p (annual, %)	3.51	1.48	1.48	1.48
κ (replacement rate)	0.40	0.40	0.185	0.40
Retirement age	65	65	65	70
Pension system (endogenous)				
N^R/N^W	0.1414	0.3064	0.3065	0.2195
τ^p (%)	5.66	12.26	5.66	8.78
$\overline{pen}$	0.475	0.569	0.303	0.580
Prices and aggregates (endogenous)				
r (annual, %)	8.58	6.82	5.89	6.82
w	0.875	0.984	1.053	0.984
K	5.16	10.04	12.82	10.96
L	6.51	9.44	10.18	10.31
Y	9.49	15.48	17.87	16.91
K/L	0.79	1.06	1.26	1.06
$5K/Y$	2.72	3.24	3.59	3.24
C/Y (%)	61.1	63.1	61.0	63.1
M/Y (%)	3.21	3.16	2.92	3.04
$5B/Y$ (%)	112	107	124	107

Notes. Each column reports a stationary general equilibrium at $\widehat{\Theta}_{\text{MD}}$, computed using the estimation grid. CF0 adjusts the pension contribution rate, CF1 reduces the replacement rate, and CF2 raises the retirement age to 70. The pension contribution rate τ^p balances the PAYG system, while government debt B adjusts to satisfy the general government budget constraint. The interest rate r is the annualized pre-tax return, and $5K/Y$ and $5B/Y$ are annual stock-to-output ratios. In CF1, the replacement rate is set to $\kappa = 0.1846$ using the PAYG financing identity, restoring the contribution rate to its baseline level (realized 5.660% against the target 5.658%). The small difference between the CF0 and CF1 dependency ratios reflects endogenous survival. Government debt is an equilibrium outcome and is not targeted in calibration.

Aggregates. Under CF0, the pension contribution rate more than doubles, from 5.66% to 12.26%. However, slower population growth also increases the relative weight of older, asset-holding households in the stationary population, contributing to capital deepening. The capital–labor ratio rises from 0.79 to 1.06, reducing the annual interest rate from 8.58% to 6.82% and increasing the wage by 12.4%. The higher wage partially offsets the increase in payroll contributions: the net-of-tax wage $w(1 - \tau^\omega - \tau^p)$ rises by approximately 3.6%. Pension benefits also increase by nearly 20%, from 0.475 to 0.569, because they are proportional to average labor income per working-age individual. Thus, capital deepening offsets much of the increase in pension contributions through higher wages, while the lower interest rate reduces the return on household savings.

Newborn welfare. Table 9 reports a birth-share-weighted consumption-equivalent welfare gain of 0.145% for CF0 relative to the 2020 baseline. The effects are small and differ in sign across demographic types, ranging from −0.31% for low-education men to +0.43% for low-education women. Under CF0, newborns face higher wages and future pension benefits, but also higher payroll contributions and a lower return on savings. These opposing effects nearly offset in expected lifetime welfare. Since newborns enter the economy without assets, they make their lifetime saving and labor-supply decisions under the new equilibrium prices. Section 5.6 examines the contributions of factor-price and fiscal changes to the aggregate welfare result. For a given capital stock, lower population growth also reduces the gross investment required to maintain a stationary equilibrium, as reflected in the investment term $(n_p + \delta)K$.

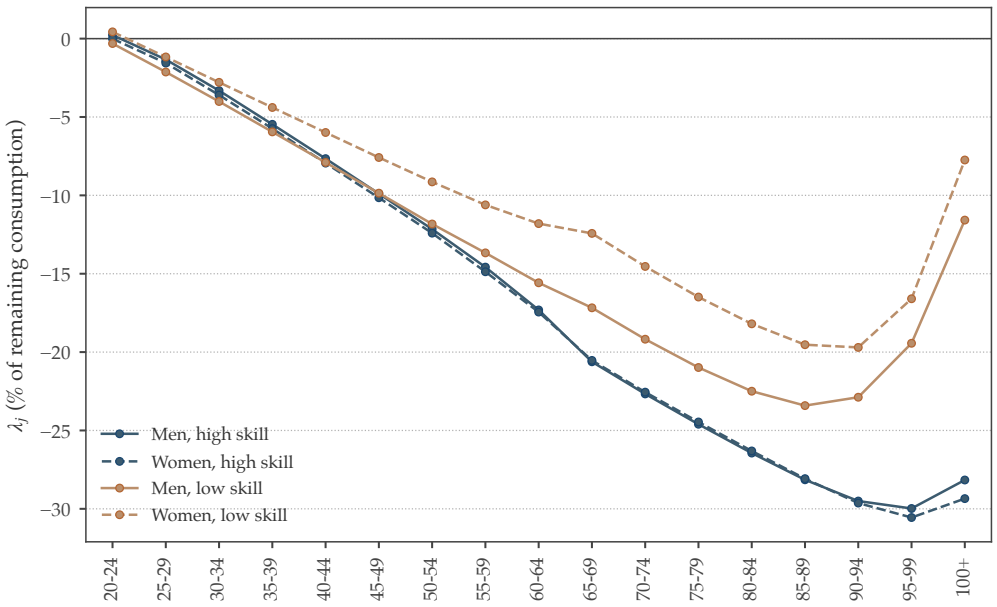
Table 9: Newborn welfare across counterfactual stationary equilibria

	CF0 vs. 2020	CF1 vs. CF0	CF2 vs. CF0	CF2 vs. CF1
Men, low education	−0.31	+8.13	+3.00	−5.39
Men, high education	+0.23	+9.30	+4.43	−5.28
Women, low education	+0.43	+7.82	+2.19	−5.82
Women, high education	−0.01	+8.58	+3.96	−4.97
Aggregate (π_1-weighted)	+0.14	+8.29	+3.11	−5.45

Notes. Entries are signed consumption equivalents, expressed as percentages of lifetime consumption and calculated according to equation (48). A positive value indicates higher newborn welfare in the first economy named in each column, while a negative value indicates lower welfare. Aggregate values are weighted by the shares of demographic types among entrants. CF0 denotes contribution-rate adjustment, CF1 benefit reduction, and CF2 retirement-age increase. All comparisons are between stationary equilibria and exclude transition effects. Relative to the 2020 baseline, CF1 generates an aggregate welfare gain of 8.41%. The aggregate CF0 estimate is approximately 0.145%, with a numerical bracketing interval of [0.1448, 0.1450].

Welfare differences by age. Figure 1 shows that the small welfare gain for newborns contrasts with increasingly negative welfare differences at older ages. Relative to the 2020 baseline, same-age residents of CF0 experience welfare losses equivalent to 1–2% of remaining consumption at ages 25–29, 6–8% at ages 40–44, and 12–21% at ages 65–69. These differences are associated with changes in the age profile of consumption. Relative to the baseline, consumption in CF0 is approximately 7% higher at ages 20–24, but 8% lower in the fifties and 20% lower at ages 65–69. Asset holdings are also lower at older ages. These patterns are consistent with the lower equilibrium interest rate shifting consumption toward younger ages. By contrast, health capital differs by less than 0.002 across the two economies. The age-specific welfare comparisons describe residents of different stationary equilibria rather than the effects of demographic aging on cohorts during a transition. Their magnitudes at advanced ages should also be interpreted in light of the consumption-equivalent properties discussed in Section 5.2.

Figure 1: Age-specific welfare differences: contribution-rate adjustment relative to the 2020 baseline



Note. The figure reports age-specific consumption equivalents, λ_j, comparing same-age residents of CF0 with those of the 2020 baseline. Negative values indicate lower welfare under CF0, expressed as a percentage of remaining lifetime consumption. High-education individuals are represented by high education and low-education individuals by low education; solid lines represent men and dashed lines women. Comparisons are between stationary equilibria and do not capture transition effects.

Welfare differences across demographic groups. The age-specific welfare losses under CF0 also differ by sex and education. Beyond age 50, losses are largest for high-education men and women, exceeding 20% of remaining consumption at ages 65–69. At these ages, the losses are approximately 17% for low-education men and 12% for low-education women. This pattern is consistent with differences in the composition of retirement resources. High-education households accumulate more assets and are therefore more exposed to the lower equilibrium return on savings. In contrast, low-education women hold fewer assets and rely more heavily on pension benefits, which increase by nearly 20% under CF0. The flat pension thus provides greater protection against the decline in returns to savings for households that depend more heavily on pension income.

The small newborn welfare differences under CF0 vary across demographic groups. Low-education women experience the largest gain (0.43%), while low-education men experience the largest loss (0.31%). These differences may reflect two features of the model. First, the increase in flat pension benefits is relatively more valuable to households with lower lifetime earnings. Second, differences in survival across sex and education affect the expected duration of pension receipt. In the baseline mortality inputs, the education gradient in survival emerges in the mid-fifties for men, whereas for women it appears only from age 70, partly because of the upper bound imposed

on baseline survival probabilities, as discussed in Section 4.6. These mechanisms may contribute to the differences in newborn welfare, but their individual contributions have not been quantified. Given the small magnitude of the welfare changes and the sensitivity of the survival probabilities to their construction, the ranking across demographic groups should be interpreted cautiously.

5.4 Benefit Reduction

Under CF1, we reduce the pension replacement rate from $\kappa = 0.40$ to $\kappa = 0.1846$, a reduction of approximately 54%, to restore the contribution rate to its 2020 level. From the PAYG financing condition (23), the required replacement rate is given by $\kappa = \tau^p_{2020}/(N^R/N^W)$. Using the CF0 dependency ratio of 0.3064 yields a replacement rate of approximately 0.1846. The resulting equilibrium contribution rate is 5.660%, compared with the baseline value of 5.658%. The small difference reflects changes in the dependency ratio arising from endogenous survival.

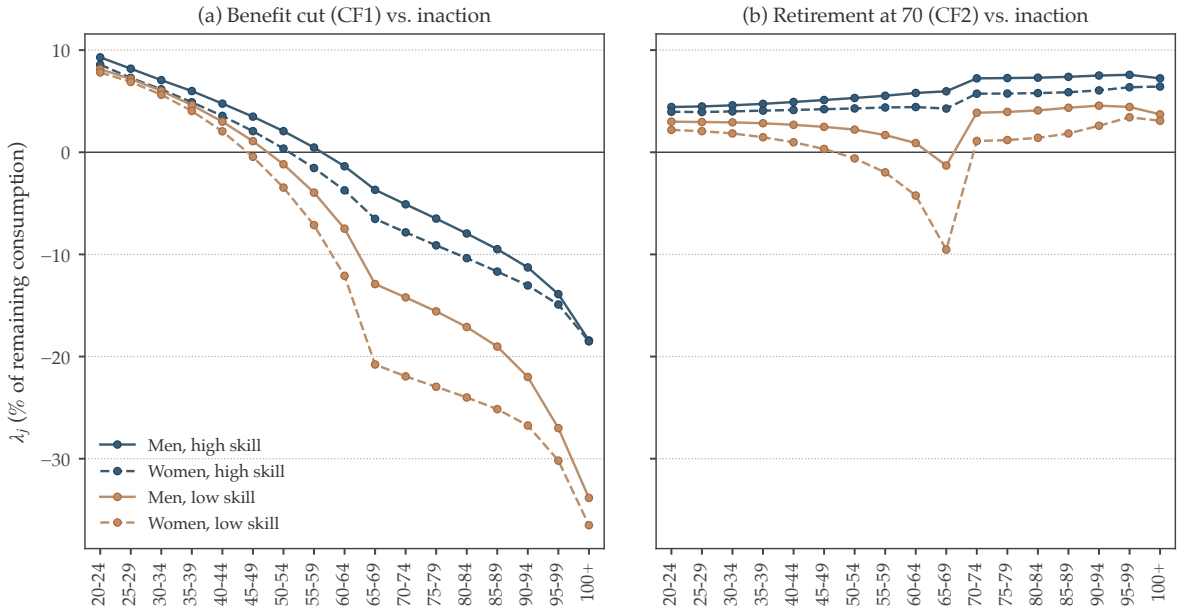
Aggregate and welfare effects. Relative to CF0, reducing the replacement rate lowers pension benefits per retiree by approximately 47%, from 0.569 to 0.303. As households rely more heavily on private savings to finance retirement, aggregate capital increases from 10.04 to 12.82. The resulting capital deepening raises the wage by 7% and reduces the annual interest rate from 6.82% to 5.89%. The combination of higher wages and lower pension contributions increases after-tax labor income, while the larger capital stock reduces the return on savings. Newborn welfare increases by 8.29% of lifetime consumption relative to CF0, with gains ranging from 7.82% to 9.30% across demographic groups. Relative to the 2020 baseline, the aggregate newborn welfare gain is 8.41%.

The welfare gains are consistent with the return differential emphasized by Aaron (1966). In a stationary PAYG system, the implicit return on contributions is associated with population growth, approximately 1.5% annually in the aged economy, compared with a market interest rate of 5.9%. Reducing the size of the pension system allows households to allocate more resources toward private savings, which offer a higher return in the calibrated economy. The resulting capital accumulation also raises wages, reinforcing the welfare gains for newborn households.

Transition effects. The 8.29% welfare gain applies to newborns in the reformed stationary equilibrium relative to CF0. It does not account for the welfare consequences of introducing the reform for generations already alive. In particular, individuals who contributed under the previous pension rules could experience substantial losses if their retirement benefits were subsequently reduced. These effects depend on the transition between equilibria, which we do not model. The steady-state welfare gains therefore do not imply that the benefit reduction constitutes a Pareto improvement.

Welfare differences by age. Panel (a) of Figure 2 shows that the welfare effects of the benefit reduction vary substantially with age. Relative to CF0, newborns and young workers experience welfare gains of 7–9% of remaining lifetime consumption, while older individuals experience increasingly large losses. At ages 65–69, these losses range from 3.7% for high-education men to 21% for low-education women. The age at which the welfare effect becomes negative also differs across groups. Low-education women experience losses beginning at ages 45–49, followed by low-education men at ages 50–54, high-education women at ages 55–59, and high-education men at ages 60–64. These differences are consistent with the relative importance of pension benefits and private savings across demographic groups. Individuals who rely more heavily on pension income are more adversely affected by the benefit reduction, while those with greater private savings benefit more from the changes in equilibrium factor prices.

Figure 2: Age-specific welfare effects of benefit reduction and later retirement



Note. The figure reports age-specific consumption equivalents, λ_j, comparing same-age residents of each reform economy with those under the contribution-rate adjustment (CF0). Panel (a) shows the benefit reduction (CF1), which lowers the replacement rate from 0.40 to 0.1846. Panel (b) shows the retirement-age increase (CF2), from 65 to 70. Positive values indicate higher welfare under the reform relative to CF0, expressed as a percentage of remaining lifetime consumption. High-education individuals are represented by high education and low-education individuals by low education; solid lines represent men and dashed lines women. These comparisons are between stationary equilibria and exclude transition effects. The productivity profiles for ages 65–69 in CF2 are estimated from individuals employed at those ages and may not represent the productivity of individuals affected by the reform.

5.5 Retirement-Age Increase

Under CF2, we increase the retirement age from 65 to 70, extending the working-age period and delaying pension eligibility by five years. The replacement rate remains at $\kappa = 0.40$. Relative to CF0, the ratio of retirees to working-age individuals falls from 0.3064 to 0.2195, reducing the pension contribution rate from 12.26% to 8.78%. Pension benefits per retiree increase slightly, from 0.569 to 0.580, reflecting an increase in average labor income per working-age individual.

Aggregate effects. Relative to CF0, raising the retirement age increases aggregate effective labor by approximately 9%, from 9.44 to 10.31. Aggregate capital increases by a similar proportion, from 10.04 to 10.96, leaving the capital–labor ratio virtually unchanged. Consequently, the equilibrium wage and interest rate remain nearly identical to those under CF0. Aggregate output increases from 15.48 to 16.91, an increase of approximately 9%. Thus, unlike the benefit reduction, which generates substantial capital deepening and changes in factor prices, the retirement-age increase primarily expands aggregate production while reducing the pension contribution rate.

Newborn welfare. Raising the retirement age to 70 increases aggregate newborn welfare by 3.11% of lifetime consumption relative to the contribution-rate adjustment (CF0), with gains ranging from 2.19% for low-education women to 4.43% for high-education men (Table 9). However, newborn welfare under CF2 is 5.45% lower than under the benefit reduction (CF1). The ranking CF1 $\succ$ CF2 $\succ$ CF0 holds for all four demographic types. Unlike CF1, the retirement-age increase does not generate substantial additional capital deepening relative to CF0, leaving equilibrium factor prices nearly unchanged. Instead, it reduces pension contributions while delaying pension eligibility and extending the ages at which individuals can work. The resulting welfare gains are positive but smaller than those obtained through benefit reduction.

Welfare differences by age. Panel (b) of Figure 2 shows that the welfare effects of raising the retirement age differ substantially across age and education groups. Relative to CF0, low-education women experience welfare losses beginning at ages 50–54, reaching 9.5% of remaining lifetime consumption at ages 65–69. Low-education men experience a smaller loss of 1.3% at ages 65–69. In contrast, high-education individuals experience welfare gains at all ages, ranging from 3.9% to 7.6%. Individuals aged 70 and older also benefit from the reform, which slightly increases pension benefits without changing their retirement status. These differences are consistent with the substantial variation in labor productivity at older ages. At ages 65–69, estimated productivity is 1.80 and 1.39 efficiency units for high-education men and women, respectively, compared with 0.88 and 0.64 for their low-education counterparts. The lower potential earnings of low-education workers make the delay in pension eligibility and the extension of

working age more costly for these groups.

5.6 Decomposition of Welfare Effects

We decompose the change in newborn expected lifetime utility in each counterfactual relative to the 2020 baseline into three components. First, we solve the household problem under the experiment's direct changes, holding factor prices (r, w), the pension contribution rate τ^p, and pension benefits $\overline{pen}$ at their baseline values. Second, we introduce the experiment's equilibrium factor prices while keeping the fiscal variables unchanged. Third, we also introduce the experiment's equilibrium contribution rate and pension benefits. The changes in lifetime utility between successive steps measure the direct household, factor-price, and fiscal components, respectively. These intermediate calculations do not impose market clearing. The three components sum to the total welfare change, but their individual contributions depend on the order in which the changes are introduced. We report the decomposition in units of expected lifetime utility rather than consumption equivalents, which are not additive across the three components.

Table 10: Decomposition of changes in newborn welfare

	Direct effect	Factor prices	Fiscal variables	Total
CF0: Contribution-rate adjustment	0	+3.247	−3.180	+0.067
CF1: Benefit reduction	0	+5.017	−1.035	+3.983
CF2: Retirement-age increase	−0.432	+3.310	−1.348	+1.530

Notes. Entries report changes in birth-share-weighted expected lifetime utility relative to the 2020 baseline, whose level is 340.60. The direct effect measures the change from the experiment's policy parameters holding factor prices and fiscal variables at their baseline values. The factor-price component introduces the experiment's equilibrium interest rate and wage, while maintaining baseline fiscal variables. The fiscal component introduces the experiment's equilibrium pension contribution rate and benefits. The components sum to the total change, up to rounding. Intermediate calculations do not impose market clearing, and the decomposition depends on the order in which equilibrium changes are introduced. Values are expressed in lifetime utility units, not consumption equivalents.

The first two rows of Table 10 show that changes in factor prices increase newborn welfare under both CF0 and CF1, but these gains are partially offset by changes in pension contributions and benefits. The direct component is zero in both experiments because neither population growth nor the replacement rate enters the household optimization problem when factor prices and fiscal variables are held fixed. Under CF0, the positive factor-price contribution of 3.247 is almost entirely offset by the negative fiscal contribution of 3.180, leaving a small net welfare gain of 0.067 utility units. Under CF1, the factor-price contribution is larger (5.017), consistent with the greater capital

deepening induced by the benefit reduction. The negative fiscal contribution is also smaller (−1.035), resulting in a net welfare gain of 3.983 utility units.

The third row of Table 10 shows that the retirement-age increase (CF2) has a negative direct welfare effect of 0.432 utility units. Unlike CF0 and CF1, this reform changes the household optimization problem directly by delaying pension eligibility and extending the ages at which individuals can supply labor. Holding factor prices and fiscal variables at their 2020 baseline values, these changes reduce newborn welfare. The factor-price component contributes 3.310 utility units, while the fiscal component reduces welfare by 1.348 units. Together, these effects yield a net welfare gain of 1.530 utility units relative to the 2020 baseline. Thus, the positive overall welfare effect arises from the combination of direct household responses and changes in equilibrium prices and pension finances, rather than from the direct adjustment to retirement age alone.

5.7 Caveats

The experiments compare stationary equilibria and do not capture transition paths or the welfare effects on generations alive when a reform is introduced. Demographic aging is implemented by changing the population growth rate n_p to reproduce the projected 2050 dependency ratio, while holding baseline survival probabilities and the composition of entering cohorts fixed. Thus, the experiments do not incorporate projected changes in mortality or educational composition. In addition, the baseline survival probabilities truncate the education gradient for women aged 50–69, as discussed in Section 4.6.

The quantitative results are also conditional on the model's fiscal structure and numerical implementation. We do not consider financing pensions through general tax revenues or government debt, which would require modifying the PAYG financing rule. Government debt instead adjusts to balance the general government budget, yielding debt ratios of 107–124% of annual output, compared with approximately 50% in Mexico. All equilibria are computed on the same numerical grid, so the reported levels are subject to numerical approximation. Finally, we do not evaluate health-policy reforms. Health capital remains close to its upper bound in the calibrated model, limiting the scope for additional health accumulation under the current specification, as discussed in Section 4.8.

6 Conclusion

This paper examines the macroeconomic and welfare consequences of demographic aging and alternative pension adjustments in Mexico using a calibrated stochastic overlapping-generations model with endogenous health and heterogeneity by sex and education. We find that demographic aging under contribution-rate adjustment (CF0) leaves aggregate newborn welfare nearly unchanged relative to the 2020 baseline.

However, comparisons between stationary equilibria reveal increasingly large welfare losses among older residents. Benefit reduction (CF1) and retirement-age increase (CF2) both improve newborn welfare relative to CF0, although their distributional consequences differ across ages and demographic groups.

Under the contribution-rate adjustment (CF0), demographic aging raises the pension contribution rate from 5.66% to 12.26%. At the same time, capital deepening increases wages and lowers the interest rate. Together with the changes in pension contributions and benefits, these general-equilibrium adjustments leave aggregate newborn welfare nearly unchanged, with a gain of 0.14% relative to the 2020 baseline. However, the welfare comparisons become increasingly negative at older ages, consistent with the greater exposure of asset-holding households to the lower return on savings.

The two pension reforms generate larger newborn welfare gains than the contribution-rate adjustment (CF0). Reducing the replacement rate from 0.40 to approximately 0.185 (CF1) restores the contribution rate to its 2020 level and increases aggregate newborn welfare by 8.29% relative to CF0. Raising the retirement age from 65 to 70 (CF2) reduces the contribution rate to 8.78% and generates a newborn welfare gain of 3.11%. All four sex-education groups rank CF1 above CF2 and CF2 above CF0. However, the age-specific welfare comparisons differ. Benefit reduction generates losses among older residents who depend more heavily on pension income, while the retirement-age increase generates losses among low-education individuals approaching retirement, particularly women. Thus, the reform that produces the largest newborn welfare gains does not necessarily improve welfare for residents at other ages.

These results are subject to several limitations. The comparisons concern alternative stationary equilibria rather than the transition to a reformed pension system, so they do not measure the welfare consequences for generations alive when a reform is introduced. The model assumes a closed economy, abstracts from informal employment, and represents the pension system as a stylized pay-as-you-go benchmark rather than the full Mexican system. The health block also reproduces only a small part of the observed education gap in survival, which we relate to the difference between medical spending in the model, chosen before the health shock, and the largely curative spending recorded in the data. Natural extensions include transition paths, an endogenous formalization margin, and a health block that distinguishes preventive from curative spending. Within these limits, the results show that pension reforms which lower long-run contribution requirements and raise newborn welfare can still impose large welfare losses on particular age and education groups. The choice among pension adjustments therefore depends on how these costs are distributed across households, and not only on the fiscal arithmetic of the pay-as-you-go system.

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Appendix A Model Appendix: Derivations and Optimality Conditions

This appendix collects two derivations behind the model of Section 3: the comparison of multiplicative and additive health amenities that motivates the specification of preferences, and the optimality conditions of the household problem.

A.1 The Amenity: Multiplicative versus Additive

Section 3.2 states that the multiplicative amenity creates an income gradient in the willingness to pay for health that an additive amenity lacks by construction. This appendix records the derivations behind that claim.

Scaling the GHH aggregator by $\Omega(h)$ makes the marginal utility of consumption depend on health,

$$u_{ch} \;=\; \Omega'(h)\left(c - v(\ell)\right)^{-\gamma} \;\geq\; 0, \tag{49}$$

with strict inequality whenever $\Xi > 0$. The sign holds at every state, since $\Omega' \geq 0$ and $(c - v(\ell))^{-\gamma} > 0$ irrespective of the bracket in (5). Health is a state rather than a within-period choice, so this complementarity reaches the medical-spending decision indirectly: through u_h in the health envelope condition (55), which supplies the marginal continuation value in (53), and through the dependence of next period's u_c on h' in the consumption Euler equation (58).

Had the amenity entered the composite additively, as $u = \left[(c + s(h) - v(\ell))^{1-\gamma} - 1\right]/(1-\gamma)$ with $s' > 0$, the same cross-partial would carry the opposite sign,

$$u_{ch} \;=\; -\gamma\, s'(h)\left(c + s(h) - v(\ell)\right)^{-\gamma-1} \;<\; 0.$$

The sign of u_{ch} is not, however, the object that governs behavior. Medical spending responds to the marginal rate of substitution between health and consumption, which is what enters the medical-spending condition (53). Under the additive form that ratio is

$$\frac{u_h}{u_c} \;=\; \frac{s'(h)\left(c + s(h) - v(\ell)\right)^{-\gamma}}{\left(c + s(h) - v(\ell)\right)^{-\gamma}} \;=\; s'(h),$$

independent of consumption. Health and consumption are perfect substitutes inside the composite, so the willingness to pay for health carries no income gradient; the negative u_{ch} above is the concavity of the outer aggregator and by itself moves nothing. Under the multiplicative form the same ratio is

$$\frac{u_h}{u_c} \;=\; \frac{\Omega'(h)}{\Omega(h)}\left[\frac{z^{1-\gamma}-1}{1-\gamma} + \chi\right] z^{\gamma}, \qquad z \equiv c - v(\ell),$$

which is increasing in c whenever $1 + \gamma\left[(z^{1-\gamma}-1)/(1-\gamma) + \chi\right] z^{\gamma-1} > 0$. The condition holds at every state at which the bracket in (5) is nonnegative, and can fail only inside the region where that bracket is negative, that is, at states the household ranks below not being alive.

The two forms also differ at the boundary of the feasible set. Because the amenity multiplies the aggregator, $u_c = \Omega(h)\,(c - v(\ell))^{-\gamma} \to \infty$ as $c \downarrow v(\ell)$, so consumption satisfies an Inada condition. Under an additive amenity the singularity sits instead at $c = v(\ell) - s(h)$, which is strictly negative whenever $s(h) > v(\ell)$. The Inada point is then unreachable, u_c stays finite as $c \to 0$, and a household can be close to indifferent between consuming and not.

A.2 Household Optimality Conditions

This appendix states the first-order, envelope, and Kuhn–Tucker conditions of the worker's problem (35) and the retiree's problem (40). Let q_j denote the multiplier on the budget constraint, and recall that the two nodes h'_k of the health lottery (16) are functions of (m, h) through $\tilde{h}$. At an optimum at which the min and max operators

in (16) are slack at both nodes, $\partial h'_k/\partial m = \partial H_j/\partial m$ and $\partial h'_k/\partial h = \partial \tilde{h}/\partial h$ for both k, with the derivatives given in (15). The interior first-order conditions are

$$\text{Consumption:} \quad u_c(c, \ell, h) = q_j(1 + \tau^c), \tag{50}$$

$$\text{Labor:} \quad -u_\ell(c, \ell, h) = q_j \, w \, \nu_j(h, \eta; g, \theta)\,(1 - \tau^\omega - \tau^p), \tag{51}$$

$$\text{Savings:} \quad q_j = \beta \sum_{k \in \{0,1\}} \Pr(k \mid j)\, \psi^{g,\theta}_{j+1}(h'_k)\, \mathbb{E}\big[V_{a,j+1}(a', h'_k, \eta'; g, \theta) \mid \eta\big], \tag{52}$$

$$\begin{aligned}\text{Medical spending:} \quad q_j(1 + \tau^m) = \beta \, \frac{\partial H_j(m, h)}{\partial m} \sum_{k \in \{0,1\}} \Pr(k \mid j) \Big\{ \big(\psi^{g,\theta}_{j+1}\big)'(h'_k)\, \mathbb{E}[V_{j+1}(a', h'_k, \eta'; g, \theta) \mid \eta] \\ + \psi^{g,\theta}_{j+1}(h'_k)\, \mathbb{E}[V_{h,j+1}(a', h'_k, \eta'; g, \theta) \mid \eta] \Big\}.\end{aligned} \tag{53}$$

Condition (53) states the two returns to medical spending that the model keeps distinct—a survival return, through $(\psi^{g,\theta}_{j+1})'$ multiplying the level of the continuation value, and a health-capital return, through $\psi^{g,\theta}_{j+1}$ multiplying its marginal value in health—each averaged over the two nodes of the health lottery. Because ς shifts the struck node down by a constant, the marginal product of spending is common to the two nodes and factors out; what differs across nodes is the point at which the survival schedule and the continuation value are evaluated. If the floor $\underline{h}$ binds at the struck node, that node's health no longer responds to m and the $k = 1$ term drops from the right-hand side of (53).

Envelope conditions. For the worker's problem ($j < j_R$),

$$V_{a,j}(a, h, \eta; g, \theta) = q_j \cdot \big(1 + r(1 - \tau^k)\big), \tag{54}$$

$$\begin{aligned}V_{h,j}(a, h, \eta; g, \theta) = u_h(c, \ell, h) + q_j \, w \, \nu_j(h, \eta; g, \theta)\, \varrho \, \ell \, (1 - \tau^\omega - \tau^p) \\ + \beta \, \frac{\partial \tilde{h}}{\partial h} \sum_{k \in \{0,1\}} \Pr(k \mid j) \Big\{ \big(\psi^{g,\theta}_{j+1}\big)'(h'_k)\, \mathbb{E}[V_{j+1}(a', h'_k, \eta'; g, \theta) \mid \eta] \\ + \psi^{g,\theta}_{j+1}(h'_k)\, \mathbb{E}[V_{h,j+1}(a', h'_k, \eta'; g, \theta) \mid \eta] \Big\},\end{aligned} \tag{55}$$

with $\partial \tilde{h}/\partial h$ from (15). The second term is the wage return to health. Its slope ϱ is common to the two skill groups, so any cross-group difference in the earnings value of health operates through the wage bill $w \, \nu_j \, \ell$—through the intercept θ and the profile $e^{g,\theta}_j$—rather than through a differential return: health is worth more, in earnings, to workers whose time is worth more. For the retiree's problem ($j \geq j_R$), $\ell = 0$ and that

term vanishes:

$$V^R_{a,j}(a,h,\eta;\,g,\theta) \;=\; q_j \cdot \big(1 + r(1-\tau^k)\big), \tag{56}$$

$$\begin{aligned} V^R_{h,j}(a,h,\eta;\,g,\theta) \;=\;& u_h(c,0,h) \\ &+\; \beta\,\frac{\partial \tilde{h}}{\partial h} \sum_{k\in\{0,1\}} \Pr(k \mid j)\Big\{\big(\psi^{g,\theta}_{j+1}\big)'(h'_k)\,\mathbb{E}[V_{j+1}(a',h'_k,\eta';\,g,\theta) \mid \eta] \\ &+\; \psi^{g,\theta}_{j+1}(h'_k)\,\mathbb{E}[V_{h,j+1}(a',h'_k,\eta';\,g,\theta) \mid \eta]\Big\}. \end{aligned} \tag{57}$$

At the terminal age $j = J$, $\psi^{g,\theta}_{J+1} \equiv 0$ kills the continuation terms; the envelopes remain well defined since $a'_J = m_J = 0$.

Euler equation. Combining (52) with (54) one period ahead and (50),

$$u_c(c,\ell,h) \;=\; \beta\,\big(1 + r(1-\tau^k)\big) \sum_{k\in\{0,1\}} \Pr(k \mid j)\; \psi^{g,\theta}_{j+1}(h'_k)\,\mathbb{E}\big[u_c(c'_k,\ell'_k,h'_k) \mid \eta\big], \tag{58}$$

where (c'_k, ℓ'_k) are next-period policies evaluated at the node-k health state.

Consumption–leisure condition. Combining (50) and (51),

$$\frac{-u_\ell(c,\ell,h)}{u_c(c,\ell,h)} \;=\; \frac{w\,\nu_j(h,\eta;g,\theta)(1-\tau^\omega-\tau^p)}{1+\tau^c}, \tag{59}$$

where the amenity cancels from the left-hand side, since $\Omega(h)$ scales u uniformly: $-u_\ell/u_c = v'(\ell)$.

Kuhn–Tucker for the borrowing constraint. The constraint $a' \geq 0$ generates a non-negative multiplier and turns the savings condition (52) into the weak inequality

$$q_j \;\geq\; \beta \sum_{k\in\{0,1\}} \Pr(k \mid j)\; \psi^{g,\theta}_{j+1}(h'_k)\,\mathbb{E}\big[V_{a,j+1}(a',h'_k,\eta';\,g,\theta) \mid \eta\big], \qquad a' \geq 0, \tag{60}$$

with complementary slackness: the product of the gap and a' is zero.

Kuhn–Tucker for medical spending. The constraint $m \geq 0$ turns (53) into the weak inequality

$$q_j(1+\tau^m) \;\geq\; \beta\,\frac{\partial H_j(m,h)}{\partial m} \sum_{k\in\{0,1\}} \Pr(k \mid j)\Big\{\big(\psi^{g,\theta}_{j+1}\big)'(h'_k)\,\mathbb{E}[V_{j+1}(\cdot) \mid \eta] + \psi^{g,\theta}_{j+1}(h'_k)\,\mathbb{E}[V_{h,j+1}(\cdot) \mid \eta]\Big\}, \tag{61}$$

for $m \geq 0$, with complementary slackness. The cap $h'_k \leq \bar{h}$ in (16) is slack wherever $\overline{H}_j m^{\zeta_h} < \bar{h}$, by the asymptote property of (14); where it binds it introduces a non-

negative multiplier $\mu_{\bar{h}} \geq 0$ with complementary slackness

$$\mu_{\bar{h}} \left(\bar{h} - h_0'\right) \;=\; 0, \qquad h_0' \leq \bar{h}, \qquad \mu_{\bar{h}} \geq 0, \tag{62}$$

entering the medical-spending condition as a subtracted term $\mu_{\bar{h}}\, \partial H_j / \partial m$ on the right-hand side of (61). At an interior optimum with $m > 0$ and both nodes strictly inside $(\underline{h}, \bar{h})$, all inequalities hold with equality and (53) is recovered.

Appendix B Estimation Details

This appendix supports the estimator of Section 4.4. It states the objective and the weighting choice, describes the algorithm that computes the equilibrium and the state space every reported number is computed on, and gives the reasons no standard errors are reported.

The estimator is deterministic. The state space is discretized and the stationary distribution over $(j, g, \theta, a, h, \eta)$ is computed by forward iteration, so every model moment is an exact distribution-weighted functional rather than a Monte Carlo average over simulated panels. The objective is therefore deterministic—re-evaluating it at the same parameter vector returns the same value—the asymptotic theory is that of classical minimum distance with no simulation-inflation term, and any non-smoothness the optimizer meets is a property of the model rather than of sampling noise. What remains is discretization: moments are kinked wherever mass crosses a grid cell.

Computing the equilibrium. The equilibrium of Definition 1 is the fixed point of three nested maps. The inner map is the household problem, solved by backward induction on the finite-horizon Bellman equations (35) and (40) together with the savings and medical-spending conditions (52)–(53) of Appendix A.2, with labor supply concentrated out through the intratemporal condition (59) and constrained maximization over the remaining controls (a', m). The middle map iterates the population measure (44) jointly with the cohort weights (2) and the health-dependent survival schedule (3) to their mutual fixed point, which is condition 10 of the definition. The outer map adjusts the capital–labor ratio until the factor-price conditions (9)–(10) and the capital-market condition (29) clear, with the contribution rate and the stationary debt level moving through (23) and (20). That outer map is not nested inside the estimation objective: parameters are obtained at fixed anchor prices, as Section 4.4 describes, and the three maps run jointly only at the estimates and in the experiments. The endogenous grid method is not available here. Medical spending enters the continuation value non-separably and through its level, since the survival derivative in (53) multiplies $\mathbb{E}[V_{j+1}]$ rather than only its marginal value, and next-period health is itself a control. The end-of-period state is therefore not exogenous and the Euler equation cannot be inverted in

closed form. At the calibrated baseline the supporting residuals are small: the largest Euler-equation residual across interior cells is $10^{-7.5}$ in marginal-utility units, the goods and capital markets clear to within 8×10^{-5} of output and capital respectively, and the cohort-share invariant implied by (3) holds to 3×10^{-15}.

Objective and weighting. The estimator solves

$$\widehat{\Theta}_{\mathrm{MD}} \;=\; \arg\min_{\Theta} \; \sum_{k=1}^{10} \left(\frac{m_k^{\mathrm{mod}}(\Theta) - m_k^{\mathrm{dat}}}{m_k^{\mathrm{dat}}} \right)^2 \;+\; \mathcal{P}(\Theta), \tag{63}$$

that is, minimum distance with $W = \mathrm{diag}\big(1/(m^{\mathrm{dat}})^2\big)$. The term $\mathcal{P}$ is a smooth penalty, zero over almost all of the admissible set, that excludes the region in which some household would assign negative value to survival; it binds nowhere near the estimates, where no household does. We do not use inverse-variance weighting, $W = \mathrm{diag}(1/\mathrm{SE}^2)$, for two reasons. The standard errors are not commensurable across this target set: one target is a convention with no sampling error at all, and the survey moments mix cluster-robust, delta-method, and bootstrap objects. And inverse-variance weighting turns the estimator into a matcher of t-statistics rather than of economic magnitudes—at the starting values it concentrated the objective on a single precisely estimated survey moment, leaving the moments that carry the paper's mechanism nearly weightless. Equal proportional weighting prices a ten-percent miss identically wherever it occurs and carries no efficiency claim. Nine parameters against ten moments over-identifies the system by one, but the over-identifying restriction is the survival-anchor overlap described in Section 4.4 rather than a restriction the theory imposes on the data, so we report no over-identification test. Nor do we claim a numerical rank condition at the estimates: two parameters sit at boundaries of the admissible set, where only one-sided variation exists. Proportional weights on two survival levels near one make the sign of their small difference nearly costless.

Inference. We report point estimates without standard errors. The numerical derivatives that the minimum-distance sandwich formula requires are not stable here—the derivative of one of the two moments that dominate the objective at the estimates is not sign-stable across candidate step sizes, so a standard error built on it would be spuriously precise—and two estimates sit at boundaries of the admissible set, where the unconstrained sandwich does not apply in any case. The nine parameters are searched under a logit transform onto their admissible intervals, so a bound is an asymptote the search approaches without attaining. The two driven to the bottom of their intervals are the amenity scale, admissible on $[10^{-4}, 5]$, and the shock arrival rate, on $[0.01, 0.436]$. A step-size study and a treatment of the two boundary estimates are prerequisites to reporting inference rather than robustness checks on it. Until then the fit of Table 5, rather than a t-statistic, is the measure of precision on offer.

Optimizer and state space. The outer search is a derivative-free simplex method terminated by a no-further-improvement rule rather than by a formal convergence criterion. All reported numbers—estimates, fit, validation, and the experiments of Section 5—are computed on the coarser of two state-space grids. That grid carries 100 asset nodes on a geometric mesh reaching 300, ten health nodes, five nodes for the persistent productivity shock, and twenty for medical spending; the finer grid raises the last three to fifteen, seven and eighty. The productivity process (13) is discretized by the Rouwenhorst method, which matches the persistence of the underlying AR(1) exactly at this node count. The two grids differ in the levels of some moments, so reported levels carry the caveat while cross-economy comparisons, which the paper's results rest on, do not. A polish on the finer grid awaits the health-block specification decision of Section 6.

Appendix C 2050 Demographic Inputs: Derivation and Sources

This appendix documents the single input that distinguishes the aged economy of Section 5 from the baseline: the 2050 old-age dependency ratio, its source, and the population growth rate that reproduces it.

Source. The demographic target for the 2050 experiments of Section 5 is the old-age dependency ratio 65+/(20–64) from CONAPO's official population projections for México (Proyecciones de la Población de México y de las Entidades Federativas, 2023 revision; CONAPO, 2023): 0.3065 in 2050, against the 2020 census value of 0.1414 that disciplines the baseline (Section 4.3). The bracketing projections (0.1746 in 2030, 0.5064 in 2070) locate 2050 on a still-steepening path. The 2050 ratio comes from the same CONAPO release as the life tables used in the first step. The UN World Population Prospects provide an alternative demographic benchmark; using them would change the target dependency ratio but not the method used to construct the stationary economy.

Method. The experiment imposes the 2050 population *structure*, not a demographic forecast path, and it does so through the single parameter the baseline already uses for the same purpose. In the stationary population, cohort weights satisfy (2), $\phi_{j+1}/\phi_j = \bar{\psi}_{j+1}/(1+n_p)$, where n_p is the five-year growth rate (the annual rates quoted here are $(1+n_p)^{1/5}-1$) and $\bar{\psi}_{j+1}$ is the distribution-weighted average survival from band j to band $j+1$—an equilibrium object, since survival is health-modulated. We hold that schedule fixed at its baseline values and bisect n_p against the CONAPO target, which gives $n^*_{2050} = 1.483\%$ per year. Because the average survivals are themselves

equilibrium objects, the bisection only proposes a growth rate; the confirming general-equilibrium solve is what we report. It realizes a dependency ratio of 0.30643, within 0.024% of the CONAPO value, and the pay-as-you-go identity evaluated at that ratio reproduces the reported contribution rate exactly.

Two arms not run. Section 5 explains why survival schedules and the birth composition stay at their 2020 values rather than moving to 2050 projections. Two further points complete that argument. A survival arm would need the CONAPO projection stripped to its exogenous component, since the model's own survival is health-modulated and a projected schedule already embeds behavior. And an education-composition arm would need an external projection of educational attainment, without which it would conflate aging with skill upgrading.

Appendix D Composition of Out-of-Pocket Medical Spending

We use ENIGH 2018 and 2024 to examine the composition of out-of-pocket medical spending across household income groups. Because ENIGH captures only between 22% and 34% of the corresponding expenditures recorded in national health accounts, we focus on spending composition rather than aggregate expenditure levels. We classify expenditures as preventive, curative, or ambiguous when their purpose cannot be determined from the survey categories. The classification differs between the two survey waves, so comparisons are made across income groups within each year rather than across years.

In 2024, 53.6% of out-of-pocket medical spending was classified as curative, 2.0% as preventive, and the remaining 44.5% as ambiguous, including over-the-counter medicines and dental and optical care. Even if all ambiguous expenditures were classified as preventive, curative care would still account for the majority of spending. Figure 3 presents the composition of medical spending across household income deciles in 2018 and 2024.

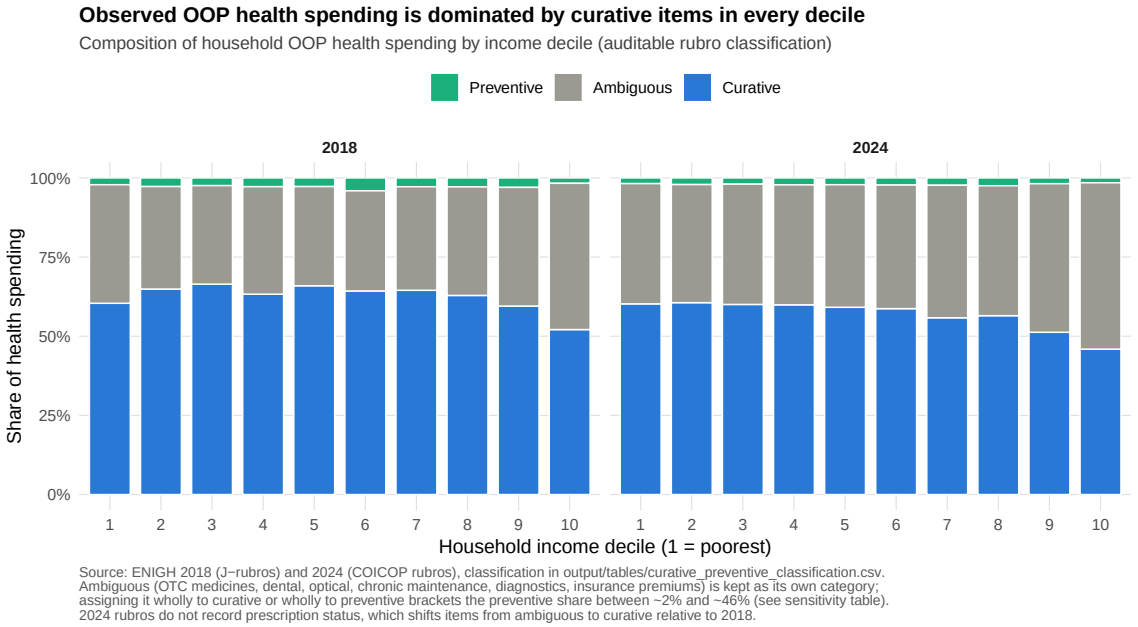


Figure 3: Composition of out-of-pocket medical spending by household income decile, ENIGH 2018 and 2024. Expenditures are classified as preventive, curative, or ambiguous. The ambiguous category includes over-the-counter medicines, dental and optical care, chronic-disease maintenance, diagnostics without a specified purpose, and insurance premiums. In 2024, curative care accounts for the majority of spending even if all ambiguous expenditures are classified as preventive. The classification uses ENIGH J-rubros in 2018 and COICOP categories in 2024, which differ in detail and do not permit direct comparisons of spending composition across years. Comparisons across income deciles within each wave remain valid. Source: ENIGH 2018 and 2024 (INEGI); authors' calculations.